REMOTE SENSING IMAGE ACQUISITION, ANALYSIS AND APPLICATIONS

Module Three

Computer-based image interpretation in practice

Remote Sensing with Imaging Radar

Voice-over script

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Lecture 1. Feature reduction

Slide 3.1.1 We now come to two practical aspects of machine learning in remote sensing. The first looks at the features we use in performing a classification, while the second addresses the very important topic of how to assess the accuracy of a thematic map.

Slide 3.1.2 As part of this work we will look at methodologies for using the various machine learning techniques we developed in module 2.

We start with looking at the topic called feature reduction. Feature reduction techniques allow us to limit the number of features to be used in a classification. Most often those features will be the spectral recordings or bands from the remote sensing instrument. But sometimes they will be spatial features, particularly if the image data has high spatial resolution.

We will not look at methods for reducing the number of spatial features, noting that convolutional neural networks do that well. Instead we will develop techniques for reducing the number of spectral features.

Slide 3.1.3 Why would we want to limit the number of features or bands?

The answer is simple. It is because the number of training samples required per class to train a machine learning algorithm is lower for fewer features. With more features, more samples per class are required for accurate training. That is a major consideration for hyperspectral image data sets. Feature reduction is carried out to avoid the so-called “curse of dimensionality” or the Hughes phenomenon when we train a classifier.

There is a second consideration: with more features, training and classification times increase, in some case more than linearly, so that classification cost also increases. Reducing features therefore reduces costs, provided the cost of the feature reduction step does not outweigh the savings gained in classification.

This raises an intriguing question. If we have to go to all the trouble of reducing the feature subset in order to develop a reliable classifier, why not just record a smaller number of bands in the first place?

Well, to answer that question we need to examine the rationale behind hyperspectral imaging.

Slide 3.1.4 Hyperspectral imaging is often referred to as **imaging spectroscopy**; indeed, that was its original name. As you probably know from your studies in science, spectroscopy means being able to identify something by reason of its spectral characteristics. Examples with which you might be familiar include mass spectroscopy, visible spectroscopy and electron spectroscopy. Sometimes the spectrum results from the absorption of radiation; sometimes it is the result of reflection or emission.

Imaging spectroscopy is a development of the familiar field of (visible) absorption spectroscopy at the pixel level. For each pixel we record the full reflectance spectrum of the substance using reflected sunlight, usually over the wavelength range of about 0.4μm to 2.5μm. We measure the sunlight reflected to the sensor on a remote sensing platform after a component of the sunlight has been absorbed by the surface material.

The reason our field is called ***imaging*** spectroscopy is because such a complete reflectance spectrum is recorded for every pixel in an image.

Slide 3.1.5 This slide summarises the essential nature of imaging spectroscopy. On the left we see an image which has been recorded in a very large number of wave bands, typically 200 or so. For a particular pixel, those bands represent samples of the reflectance spectrum of the corresponding region on the ground—or more correctly, the reflectance spectrum of the **surface material corresponding to the pixel**.

From those samples we can reconstruct the reflectance spectrum of the pixel as shown on the right-hand side. Notice that the reconstructed spectrum, particularly if there are sufficient samples, shows various important diagnostic features such as the dip corresponding to chlorophyll absorption in the red region and the water absorption bands. There are other finer absorption features too, especially for soils and minerals. They do not show up on this vegetation example.

Slide 3.1.6 One of the great benefits of recording the full reflectance spectrum for a pixel is that we can use scientific knowledge to identify the pixel rather than depend on supervised learning and machine classification. Expert spectroscopists can undertake that analysis from their knowledge of the spectral response properties of materials. While that is a viable approach, most often in hyperspectral remote sensing the recorded reflectance spectrum of the pixel is compared with a library of pre-recorded spectra as in the next slide.

Slide 3.1.7 This slide illustrates the approach to image interpretation based on library searching. The image captured by the sensor consists of pixels, for each of which a full reflectance spectrum has been recorded. That spectrum is then compared against a library of spectra previously recorded in the laboratory, allowing a label to be attached to the pixel. Often that can be facilitated by using expert system techniques, in which the knowledge of an expert analyst is encoded in sets of rules that are applied to the data.

Slide 3.1.8 Although spectroscopic analysis is regularly applied in the earth sciences using recorded hyperspectral image data, in many applications we still wish to use machine learning methods for labelling a pixel. This approach is convenient and does not rely on expert knowledge or recorded spectral libraries.

But we then have to face the problem that there are too many bands recorded to allow all of them to be used as features in a supervised classification exercise. That is because it is too difficult to obtain enough training data per class to allow our classifier parameters to be estimated reliably.

We now therefore need to find methods that will allow us to identify a subset of the recorded bands that is still sufficient for the development of an accurate classifier.

Slide 3.1.9 Before proceeding there is some nomenclature that we should keep in mind. First, "**features**" is the name given to the input measurements to our various classification algorithms—most usually they are just the recorded bands, or some transformed version of the bands after, say, the application of the principal components transform. In object detection they may be spatial descriptors. In some applications the features sets may include both spectral and spatial descriptors.

We are now embarking on a search for methods to reduce the number of features to use, not unreasonably called **feature reduction** in general.

One approach to feature reduction is to select subsets, which still function effectively. That is called **feature selection**. Feature selection is the most common form of feature reduction, but other approaches also exist as we will see in the following material.

Slide 3.1.10 Feature reduction has been a focus in the machine learning world ever since the first algorithms were developed. Many techniques have been proposed over the years in remote sensing, especially since hyperspectral image data became available.

In these lectures we're going to look at some of the more common feature reduction techniques. For those who wish to take this study further the paper referenced on this slide is a comprehensive overview of the field.

Slide 3.1.11 Essentially, we are going to look at three different approaches.

The first will depend upon an analysis of the correlations among the recorded bands of data. We will see that those correlations allow us to consider groups of bands, and will let us ignore correlations from group to group.

The second approach involves transforming the spectral data in such a way that we can ignore the least significant transformed bands.

The third approach we will look at involves measures that guide us in discarding some of the original bands of data without significantly compromising the quality of classification. These techniques fall into two groups: one assumes that the classes can be described by probability distributions, while the other does not require any such assumptions.

Slide 3.1.12 Let's take stock of what we have said so far before we embark upon the actual procedures for feature reduction.

Hyperspectral data can have as many as 200 or so bands.

The field of imaging spectroscopy depends on reconstructing the reflectance spectrum for a given pixel from the set of hyperspectral measurements.

Often the reflectance spectrum is identified by reference to a library of prototype spectra. It is important, for that to be effective, that the recorded spectrum be radiometrically calibrated.

Most of our focus will be on analysis by machine learning methods. To be effective they require the number of features to be reduced. We will look at feature reduction methods.

Slide 3.1.13 In the second question here pay particular attention to the correlations of each of the four bands with all three of the other bands.

Lecture 2. Exploiting the structure of the covariance matrix

Slide 3.2.1 In this lecture we are going to examine the structure of the covariance matrix and thus the correlation matrix to see how their properties help us undertake feature reduction.

Slide 3.2.2 We have met the covariance matrix several times in the past. It is the starting point for the principal components transformation and, along with the mean vector, defines the class signature when we undertake maximum likelihood classification.

In the latter context we know that the accurate computation of the class covariance matrix can be a problem for hyperspectral data if we do not have enough training samples per class. Remember, we have to have enough independent training samples in each class in order to estimate reliably the elements of the covariance matrix. Generally, that is not a problem with the principal components transformation because, then, the covariance matrix is computed using all of the available training samples and not just those for an individual class.

So, we can examine the covariance matrix for the data as a whole; we will do that now and note that it has some interesting structural properties. Rather than the covariance matrix itself we will look at the correlation matrix instead which, remember, is derived from the elements of the covariance matrix.

Slide 3.2.3 Let's look at an example of a real correlation matrix. Here we will examine the matrix for the Jasper Ridge USA image, which has 196 bands. The correlation matrix for this image will be of dimensions 196 x 196. This is too large for us to write down. We can, however, represent it in an image form as shown to the right. Remember, correlations are in the range of -1 to + 1. A correlation of zero means just that—there is no correlation between the pair of bands, whereas a correlation of +1 means total positive correlation, while -1 means total negative correlation. If we choose a greyscale where black represents zero correlation and white means total positive or negative correlation, we get a very interesting representation of the correlation matrix.

Note that there are regions of high correlation distributed down the principal diagonal of matrix. That means that the bands in those blocks are highly correlated. Where there are black blocks the associated bands have little correlation among them. There are some groups of bands which lead to correlations off the diagonal suggesting some inter-relationships of bands in the middle infrared and between the middle infrared and visible regions.

In general, though, we can assume that most of the correlation activity is down the principal diagonal. What does that now suggest for feature reduction?

Slide 3.2.4 Here we neglect the off-diagonal correlations and represent the correlation matrix just by the blocks that form down the diagonal. The matrix now becomes a block diagonal matrix, which has a number of very interesting properties.

If we write the block diagonal covariance matrix in the form shown in the centre of this slide, composed of H separate blocks down the diagonal, a number of simple and useful properties arise, as we now see on the next slide.

Slide 3.2.5 First, the determinant of a block diagonal matrix is the product of the determinants of the individual blocks. That means the logarithm of the determinant is the sum of the logarithms of the individual determinants.

Secondly, the trace of the determinant is the sum of the traces of the individual blocks.

Finally, the inverse of the correlation matrix can be computed using the inverses of the individual blocks, as shown on the bottom of this slide.

Slide 3.2.6 Now what happens to a column vector which has the same dimensions as the covariance or correlation matrices? This could be, for example, the mean vector which corresponds to the covariance matrix. Clearly it can be represented as the concatenation vertically of a set of smaller vectors corresponding to the blocks of the covariance matrix.

That leads to a particularly important matrix identity which says that the vector-matrix-vector construct we use regularly in image processing can be decomposed into the sum of the same expression computed over each of the blocks.

We now have all the material needed to show how block diagonalizing the covariance matrix simplifies the principal components transformation and the maximum likelihood procedure.

Slide 3.2.7 Just before we do that, we can note a further simplification.

Rather than utilizing the actual blocks of high correlation down the diagonal, we could just restrict our attention to the correlations that exist among immediately adjacent bands. For example, as shown in this slide, we could just use the 2 x 2 or 3 x 3 blocks indicated. While probably not as accurate as retaining the true block diagonal structure of the matrix, such a simplification reduces dramatically the number of bands that have to be considered when training a classifier, as we will see shortly.

Slide 3.2.8 Irrespective of whether we use that most recent simplification, or stay with the original block diagonal form of the matrix, we now look at how the block structuring of the covariance matrix affects the formula for the principal components transformation.

Remember, to compute the eigenvalues of the covariance matrix we have to solve the characteristic equation shown here in determinant form. When the covariance matrix is block diagonalized, the characteristic equation is the product of a set of equations of smaller dimensions. Thus, the roots of the equation are the eigenvalues of the component block matrices. Likewise, we can find the eigenvectors and the principal components transformation matrix by computing everything on a block basis and then combining the results.

Slide 3.2.9 Now let's see how block diagonalizing the covariance matrix (and equivalently partitioning the mean vector) affects the maximum likelihood classifier.

Using the rules above for block diagonal matrices, the discriminant function for the Gaussian maximum likelihood classifier can be shown to reduce to the expression in the middle of this slide.

Although using the block diagonal approximation to the covariance matrix ignores correlations among bands that are widely separated, it does have significant benefits in terms of reducing the dimensionality that has to be considered when applying the maximum likelihood classification rule.

The largest matrix for which the elements need to be estimated using training data is that of **the largest block**. In general, that will be significantly smaller than the original matrix, meaning that many fewer training pixels are needed for reliable estimates, rendering the maximum likelihood rule useful for hyperspectral thematic mapping.

Slide 3.2.10 We now look at an example of the performance of the maximum likelihood classifier with block diagonalization, taken from the paper listed.

Two AVIRIS images were used, based on the five different tests listed at the bottom of the slide. Note that the last test applies just the minimum distance rule.

Full details will be seen in the paper referenced.

Slide 3.2.11 The results of the tests are shown here in graphical form, so they can be compared. The classification accuracy shown is the average performance on training and testing data. Note how good the results are, particularly for the actual diagonal blocks.

What these results don't show, but which is included in the original paper, is that there is a significant improvement in classification time with the smaller blocks. That is because the classification time for the maximum likelihood rule is quadratically dependent on the number of bands used

Slide 3.2.12 There are four key messages from this lecture:

- Covariance and correlation matrices can be represented as images
- Strong band-to-band correlations appear in blocks, largely down the principal diagonal of the covariance (or correlation) matrix
- The block diagonal structure can be exploited to simplify image processing operations that require computation of a covariance matrix
- The block diagonal form of the covariance matrix allows the maximum likelihood classifier to be applied to hyperspectral image data

Slide 3.2.13 The last two questions here focus your attention on understanding what the image representation of the correlation (or covariance) matrix tells us.

Lecture 3. Feature reduction by transformation

Slide 3.3.1 In this lecture we are going to look at what can be done when using data transformation as a feature reduction tool. This has been a common approach for many years, particularly using the principal components transform.

Slide 3.3.2 Remember that the principal components transformation maps image data into a new, uncorrelated coordinate system or vector space.

It produces a data representation in which the most variance is along the first axis, the next largest variance is along a second mutually orthogonal axis, and so on.

The later principal components would be expected, in general, to show little variance. They could be assumed to contribute little value to separability and thus could be ignored, thereby reducing the essential dimensionality of the vector space, and leading to more effective classification.

As noted earlier, this is a widely used approach. It requires the computation of a covariance matrix, but using the data set as a whole, so there is generally sufficient training data available to be able to estimate the global covariance matrix accurately.

We will illustrate the approach by working through an example.

Slide 3.3.3 Consider the two class, two-dimensional example shown here.

If you look at the distribution of the data by class you can see, even now, that the two classes should be separable in an axis aligned at about 50 degrees to the original axes—in other words the first principal component. Now let's verify that.

Using the standard formulas, the global mean vector and covariance matrix are as shown here. We then compute the eigenvalues of the covariance matrix, and thus the corresponding eigenvectors. Note that the eigenvalues suggest that a large part of the data variance will be along the first principal component.

Slide 3.3.4 From the eigenvectors we get the principal components transformation matrix, from which we generate the formula for the first principal component as a function of the original axes. That is shown plotted in the diagram, along with the projections (components) of the data points on to that axis.

As expected, the classes can be separated in the first principal axis, meaning the second axis can be ignored. Subsequent classification can, therefore, be based just on the first PC alone. Thus, in this simple example, the number of features has been reduced from two to one.

Slide 3.3.5 There are some situations, however, where the principal components transformation will not work as a feature reduction tool, such as that illustrated here.

Feature reduction using the principal components transform only works if the classes are separable along a reduced set of (principal) axes, as in the example just considered. In the example here, we could not use principal components analysis for feature reduction because of the manner in which the classes are distributed. There is, however, a transformation that can be used in such a situation—it is called **canonical analysis**.

Slide 3.3.6 If we could find an axis, such as that shown in the bottom diagram then, again, we could represent the data with a single feature, rather than the two original bands. That axis is defined such that when the classes are projected on to it, they appear as far apart from each other as possible, while having the smallest spread.

If, along that preferred axis, we define the "among class variance" and the average "within class variance" as shown in this slide then we want to find the axis rotation such that we maximise the ratio of those two variances.

In order to find this expression and the actual axis transformation we have to express the problem in matrix form, so that it applies to any number of features. We need also to express it in terms of the variances in the original features.

Slide 3.3.7 In this slide we define the key properties of importance when finding the so-called canonical axes. They include the global mean, and the two variances, expressed in matrix form, as required.

Slide 3.3.8 This looks a bit complicated, but it follows an approach almost identical to that of the principal components transform.

Let $\mathbf{y} = \mathbf{D}^{\mathrm{T}}\mathbf{x}$ be the required transform that generates a new set of axes $\mathbf{y}$ in which the classes have optimal separation. By the same procedure that was used for the principal components transformation we can show that the within class and among class covariance matrices in the new $\mathbf{y}$ coordinate system can be expressed in terms of those in the original $\mathbf{x}$ coordinates, as shown in the centre of this slide. As with PCs, the row vectors of $\mathbf{D}^{\mathrm{T}}$ define the axis directions in $\mathbf{y}$ space.

Let $\mathbf{d}^{\mathrm{T}}$ be one particular vector, say the one that defines the first axis along which the classes will be optimally separated. Then the corresponding within class and among class variances will be as shown on the bottom of the slide.

Slide 3.3.9 We now set up the measure shown as equation (A), which we have to maximise (differentiate) with respect to the axis direction $\mathbf{d}$. We then solve the generalized eigenvalue equation shown, subject to the normalizing constraint at the bottom. That solution generates the canonical axes for us, as we see in the analysis on the next slide.

Slide 3.3.10 Here we see a two class data set, which is not separable in the first principal components axis, but will be seen to be separable in the first canonical axis, using the computations shown on this slide for the covariances.

Slide 3.3.11 Substituting the covariance calculations into the characteristic equation at the top of this slide yields the eigenvalues of 2.54 and 0. Having a zero eigenvalue tells us that there is only one canonical axis in this example.

Using the non-zero eigenvalue in the characteristic equation at the top, along with the normalizing condition, leads us to the new axis shown in the equation at the bottom of the slide. That axis is plotted on the original diagram, in which it is seen that the two classes are now separable using just one (transformed) feature.

Slide 3.3.12

- Feature reduction can be carried out by transforming the original spectral bands to a new coordinate system in which some transformed bands can be discarded without affecting the separability of the classes.

- The principal components transform is the most popular method to use. Even though it is based on the whole data set—the global mean vector and global covariance matrix—provided the classes themselves are approximately distributed along the first few principal axes, it shows good results.

- If the classes do not distribute along the principal axes, then principal components analyses is not a good technique to use for feature reduction. It is better then to use the transformation associated with canonical analysis.

- Both principal components and canonical analysis depend on covariance information. In the former it is a global measure and thus easier to estimate accurately with limited training data, whereas for canonical analysis class-specific covariances are required and thus the method has limitations with high dimensional data such as hyperspectral imagery

Slide 3.3.13

These questions direct your attention to an alternative measure for computing class separability.

Lecture 4. Separability measures

Slide 3.4.1 This lecture treats the third approach to feature selection, in which we set up measures to assess the importance to a classification exercise of the original sets of bands.

Slide 3.4.2 This is the more classical approach. It entails setting up measures that help us to evaluate the relative importance of each of the bands, allowing us to discard some when we want to undertake thematic mapping. Generally, these approaches depend upon a measure known as **separability**, which can be based on

- an assumption that the classes of interest can be described by probability distributions or
- avoiding the use of distributions.

Obviously, the first method is restricted to image data sets with small numbers of bands because of the need to estimate the statistics of the probability distributions—such as the Gaussian model that is almost always used. It is the old problem of not having enough samples per class in order to estimate the distribution parameters reliably. Provided the number of bands does not exceed about 10, those methods should be suitable.

The second set of methods was developed to help undertake feature reduction with high dimensionality data sets, such as hyperspectral imagery. They are generally more complicated to develop but are more widely applicable.

Slide 3.4.3 Measures of separability tell us how distinct or spectrally different **two** thematic classes are.

Remember, classes are defined by sets of features. Our objective here is to see whether we can use fewer features and yet still carry out an acceptable classification.

We use separability to tell us whether classes based on chosen subsets of features are distinct enough to allow good classification results to be achieved.

Generally we adopt a procedure such as:

- We start with the full feature subset and, via a separability metric, compute how distinct the classes are
- We then remove a feature (or sometimes sets of features) and see by how much separability has been reduced
- If separability is not badly affected, we could leave that feature out; otherwise we keep it and test other features

We then benefit from reduced classification cost and avoidance of the Hughes phenomenon, but the search is an exhaustive one, requiring all features to be assessed for a full evaluation.

Slide 3.4.4 Divergence is one of the earliest and simplest measures of separability. It is based on quantifying the overlap of two class distribution functions, as illustrated in the diagram on this slide.

Suppose we have two classes ω_1 and ω_2 that are separable in the two features x_1 and x_2. Being separable means they can be easily mapped as different classes by a classification algorithm, leading to a high classification accuracy.

If we discard feature x_2 and retain only x_1, as shown on the bottom of the diagram, we see that the classes then overlap and thus cannot easily be separated with high accuracy by a classifier using just that feature alone, in place of the two original features (bands).

Divergence tells us the penalty we pay by way of loss of separation if we reduce the number of features by discarding one or more.

Slide 3.4.5 Without going into the derivation, the formula here shows the computation of divergence for a pair of classes assumed to be modelled by Gaussian, or multidimensional normal distributions.

As we noted on the previous slide, we use divergence to see how the similarity of classes is affected if we remove bands. What we are looking for is whether the separability of the two distributions is made noticeably worse by the removal of features. If it is, then those features are important when running a classification and should not be ignored. On the other hand, if removing a feature doesn't drop the separability by very much, then it could be removed when doing a classification, without significantly affecting accuracy.

Divergence has a number of important properties:

1. It is always positive
2. It is zero if the distributions are identical
3. It never decreases with the addition of features. That means it is never higher if features are removed.

Slide 3.4.6 Another measure of separability is the Jeffries-Matusita distance, which is again a measure of the average distance between two distributions i, j. For normally distributed classes it has the form shown, involving an exponential function of a distance measure, called the Bhattacharyya distance. The exponential term will be seen soon to be very important.

While JM distance bears some similarity to divergence, its significant difference is the exponential term. To see the effect of that we sketch on the next slide how divergence and JM distance would vary as a function of the distance between two distributions (measured by the difference in mean vectors).

Slide 3.4.7 By looking at the formulas for divergence and JM distance, the behaviours shown on the two diagrams shown here can be seen.

Divergence increase without bound as the distance between classes increases, whereas the JM distance curve saturates. Which behaviour is the more useful?

If we think about how the probability of correct classification might change as the two classes move further apart, it is more likely to behave like the JM curve, as we will see on the next slide.

That is because the overlap between distributions drops quickly at first and then approaches zero. Once the distributions are well separated (and thus classification accuracy is close to 100%), further separation will have little effect on the classification result.

Slide 3.4.8 Here we see the probability of correct classification compared with the behaviours of divergence and JM distance. This suggests that the JM curve is more realistic, all because of the saturating behavior of the exponent in its formula.

However, because JM distance requires the computation of matrix inverses of the **sums of all pairs** of classes, it is time consuming if it is to be used to check the average similarity of many spectral or information classes.

By comparison, divergence is much quicker to compute because the inverse of each class covariance matrix is computed only once, even when checking the average similarity of a set of classes.

Slide 3.4.9 A third separability measure was developed heuristically to take advantage of the inherent benefits of both divergence and JM distance. Called Transformed Divergence it embeds the usual divergence expression into a saturating exponential function, so that its overall behavior emulates that of the probability of correct classification. It is also faster to compute than the JM distance.

Transformed divergence is perhaps the most commonly used measure of separability in remote sensing when classes can be described by the normal probability distribution. As we noted earlier, that generally applies when the number of bands or features does not exceed about ten.

Slide 3.4.10 One of the important uses of separability measures is to be able to assess beforehand whether the spectral classes are well enough separated to ensure a certain level of classifier accuracy.

While the actual performance of a classifier depends on many factors, including how well the analyst has defined the spectral classes, collected reference data and trained the algorithm, if the classes are not well separated then there is an upper limit on the classifier accuracy that can be achieved.

The formula shown on this slide provides an upper limit on the probability of correct classification, in a two class case, as a function of transformed divergence. This is plotted on the next slide.

Slide 4.3.11 Here we see the theoretical upper bound of classification accuracy versus transformed divergence, along with some empirical classification results computed from 2790 trials. As seen, unless transformed divergence is close to its maximum, binary classification accuracy falls well short of the theoretical upper limit.

Slide 3.4.12 We met clustering algorithms in the last module; they are the basis of unsupervised classification and form a component of a combined unsupervised/supervised classification methodology that we will visit shortly.

One of the last stages in clustering is to evaluate the size and relative locations of the clusters produced. If clusters are too close to each other in spectral space they should be merged. What does "too close" mean?

If cluster mean and covariance data is available, we can make merging decisions based on a pre-specified transformed divergence. By establishing a desired accuracy level for the subsequent classification, from which the corresponding value of transformed divergence can be specified, cluster pairs with separabilities below that value should be merged.

Slide 3.4.13 That completes our treatment of measures of separability based on distribution functions. In the next lecture we will relax that requirement and consider separability measures that can be used when it is not possible to estimate second order parameters such as covariances; that is usually the case with hyperspectral data.

Slide 3.4.14

- For any classifier it is good to use as few features as possible in order to constrain classification time, and thus costs

- Feature reduction is important in maximum likelihood classification in order to avoid the Hughes phenomenon, in which not enough training samples are available to develop reliable estimates of the elements of the covariance matrix

- Divergence, Jeffries-Matusita (JM) Distance and Transformed Divergence are three methods commonly used when classes can be modelled by probability distributions

- JM distance performs well but is time consuming to evaluate, whereas divergence is faster to use but doesn't work well when classes are widely separated in spectral space. Transformed Divergence combines the benefits of both approaches.

- Separability measures allow us to place an upper bound on classification accuracy

Slide 3.4.15 Remember to consult the solutions if you are having problems with any of the quiz questions at the end of each lecture.

Lecture 5. Distribution-free separability measures

Slide 3.5.1 We now look at separability measures that can be used when it is not possible to represent classes by probability distributions. We will look at two different measures. Some of this material is a bit conceptually.

Slide 3.5.2 The reason we need to develop separability measures that can be used without class models is so we can handle data of hyperspectral dimensionality; in such a case there are insufficient training samples for the generation of reliable covariance matrices.

Separability measures are still possible in these circumstances, but they are a bit more complicated to develop, since we cannot rely on measures that look at the difference between, say, two normal distributions.

Slide 3.5.3 **ReliefF** is one of the commonly used measures for feature selection—that is for evaluating existing features and seeing which of them could be discarded.

This measure gives features a weight, which is adjusted by reference to the classes of data being used. Those features with weights below a user-specified threshold are discarded.

We will develop it by reference to the two class, two-dimensional data set shown on the slide. The classes have been drawn intentionally so that one feature x_1 does not aid separation while the other x_2 does.

The process commences by selecting a pixel at random from one of the classes. We then find its nearest neighbours in the same **and** the other class, as shown.

We now want to derive a measure that gives more weight to feature x_2 than feature x_1 with respect to those chosen pixels.

Slide 3.5.4 We have two features x_1 and x_2. We now define a weight **for each feature** that tells us how important it is with regard to class separation. Call these weights ω_1 and ω_2 respectively.

The weights are initially set at zero and then updated using the pixel we have chosen randomly. We will shortly choose further random pixels, to give us a better measure of the weights, but for the moment just concentrate on the one in the diagram.

For the weight corresponding to the i^{th} feature we use the updating rule shown.

In that rule, x_i is the feature of the randomly chosen pixel, x_i^s is the corresponding feature of the nearest neighbour from the same class and x_i^o is the corresponding feature of the nearest neighbour from the other class. d is a distance measure.

Slide 3.5.5 Applying the updating rule to the diagram shown, we see that the adjustment will be small and negative for feature x_1 but large and positive for feature x_2. That means the weight for the second feature increases, indicating its relative importance, while that for the first feature drops.

The same process is carried out several times, using a set of $\boldsymbol{m}$ **randomly chosen sample pixels**, updating the weights each time. The distance values are normalised by the number of samples m.

If there is a reasonable distribution of pixels in each class, a little thought will show that the weight for feature x_2 will go on increasing relative to that for x_1.

Slide 3.5.6 A helpful modification is to use a set of the K nearest neighbours (in each class) to the randomly chosen pixels, as indicated in the diagram. In this case the updating rule becomes as shown by the formula, which indicates normalisation by both the number of trials and the number of nearest neighbours.

Again, the process is initialised with all weights ω_i set to zero. Each weight is then updated by selecting m random samples (pixels) using the above rule. At the completion of the process those features x_i with weight values above a threshold are kept for subsequent quantitative analysis (classification).

Slide 3.5.7 The reliefF method, as originally formulated, applies to two classes. It is readily extended to the more usual multi-class situation by adding an "other class" term for each of the classes other than that from which the random sample is taken, leading to the formula shown on this slide.

The x_i^{ok} are now the i^{th} feature of the k^{th} nearest neighbour of the current random sample in each of the other classes. The probability expression $\frac{p(o)}{1-p(s)}$ weights the contributions from the other classes in proportion to their (and the same class) prior probabilities.

In all these formulas the distance measure can be any convenient metric. Euclidean and city block distance are the most commonly used.

Slide 3.5.8 Another feature selection technique that avoids the need to use probability distribution models for classes is **non-parametric discriminant analysis** (NDA). Like canonical analysis, it sets up measures of the distribution of pixel vectors within classes, and the dispersion of the classes themselves in spectral space, and then seeks an axis transformation which provides maximum separation between the classes based on those measures.

Instead of using covariance matrices, though, it employs the concept of scattering matrices seen in the following slide.

This is a bit complicated conceptually, but not hard mathematically. If you find the concepts difficult it will affect your understanding of the equations. Since this procedure is only one of a number of distribution free methods, you may wish to pass over the material if you find it too complicated. But if you wish to employ it in practice, software is available that will guide you through its use.

In its simplest form, NDA examines the relationship between the training pixels of one class and their nearest neighbour training pixels from another class.

For example, let $\boldsymbol{x}_{j\in s,NNi\in r}$ represent pixel j from class s that is the nearest neighbour of pixel i from class r as shown in the diagram.

Slide 3.5.9 We can describe the distribution of class r pixels with respect to their nearest neighbours in class s by a covariance-like calculation. However, because we are now not describing the distribution of pixels about a class mean (a parametric description), it is better to use a different term than covariance matrix. To talk about the scatter of pixels with respect to each other we use the term *scatter matrix*.

The scatter of all of the training pixels from class r about their nearest neighbours from class s is defined by the scattering matrix definition shown in the equation, which computes the expected value of the distance between those pixels, times the transpose of that distance, giving a matrix, effectively of the distance squared.

In this expression $\boldsymbol{x}_{i\in r}$ is the i^{th} pixel from class r and the $|\omega_r$ conditionality reminds us that the calculation is determined by pixels from class r.

Slide 3.5.10 We then do a similar calculation for the scatter of the training pixels from class s about their class r nearest neighbours, and then average the two measures. Usually the average is weighted by the prior probabilities, or relative abundances, of the classes, as shown in the first formula on the bottom of this slide.

Often NDA uses not just the nearest *neighbour* but instead a set of k class s training pixels as a nearest *neighbourhood* for each class r training pixel. The local mean over that neighbourhood is then used in the calculation of the between class scattering matrix, giving rise to the second formula for the between class scatter matrix, which includes a mean vector term for the k nearest neighbours in class s which is shown on the next slide.

Slide 3.5.11 The mean vector in the previous calculations is given by the expression shown here. Note that if k, the size of the neighbourhood, is the same as the total number of training pixels available in class s then the “ local” mean becomes the class mean, and the between class scatter matrices resemble covariance matrices, although taken around the mean of the opposite class rather than the mean of the same class.

Slide 3.5.12 The approach has to be made applicable to a multi-class situation. In doing so we note that there are as many weighted means of the pixels "from the other class" as there are "other classes." This is illustrated in the diagram for the case of three classes: r, s and t. It is easier to express the expectations in algebraic form, so that for C total classes the among-class scatter matrix is as in the formula given.

In this expression the inner sum computes the expected scatter between the N_r training pixels from class r and the mean of the nearest neighbours in class c (different for each training pixel); the middle sum changes the class (c), still relating to the training pixels from class r; the outer sum changes the class (r) for which the training pixels are being considered. The latter computation is weighted by the prior probability for the class.

Slide 3.5.13 Having derived an expression for how the classes are scattered with respect to each other, we need to look at how the pixels scatter **within** the classes. We define the mean vector of a class based on the k nearest neighbours of the i^{th} pixel in class r from the same class as shown, which leads to the within-class scatter matrix in the second expression.

Slide 3.5.14 How do we now use the two scatter matrices $\boldsymbol{S}_W$ and $\boldsymbol{S}_A$ to carry out feature reduction?

Remember, we are looking for a new set of axes in which $\boldsymbol{S}_A$ looks as large as possible while, at the same time, $\boldsymbol{S}_W$ looks as small as possible. We define the axis transformation in the usual way by the matrix equation

$$\mathbf{y} = \mathbf{D}^{\mathrm{T}}\mathbf{x}$$

and look for a $\mathbf{y}$ coordinate system in which we can maximise the joint measure

$$J = \mathrm{tr}\{\boldsymbol{C}_{W,\boldsymbol{y}}^{-1}\boldsymbol{C}_{A,\boldsymbol{y}}\}$$

Once the new axes have been found we will have maximum class separation in the first, followed by the next best separation in the second, and so on. We then retain a subset of those transformed bands (axes) as the feature-reduced representation of the data we wish to classify.

Slide 3.5.15

- For high dimensional imagery we need measures of separability that do not depend on the need to compute a covariance matrix

- The method known as ReliefF devises a weighting scheme. Each feature (band) is allocated a weight to indicate its significance. Bands with weights less than a user-specified threshold can be discarded when undertaking a classification.

- Non-parametric Discriminant Analysis (NDA), parallels canonical analysis in that it seeks a new set of axes in which to represent the pixel vectors, such that classes have maximum separation in the first new axis, next best separation in the next axis, and so on. Non-parametric scatter matrices are used in the calculations. After transformation, those low order features which do not help class separation are discarded, resulting in a data set with smaller dimensionality

Slide 3.5.16 Again, please consult the solutions if you need to, in order to help consolidate your understanding of distribution-free feature reduction techniques.

Lecture 6. Assessing classifier performance and map errors

Slide 3.6.1 With this set of lectures on accuracy assessment we come to the end of our material on general remote sensing, before then completing the course with a set of lectures on remote sensing with imaging radar.

Slide 3.6.2 Assessing how a classifier performs and the accuracy of the thematic map generated is one of the most important topics in operational thematic mapping in remote sensing.

We will see that classifier performance and map accuracy are not the same thing.

This is a crucial matter in practical remote sensing. If significant economic decisions are going to be based on the thematic maps and class hectarages, generated by the machine learning techniques we have covered in this course, then we have to be as sure as we can that we are interpreting the results correctly.

In this series of lectures we will look at

- How we assess the performance of a classifier
- The relationship between classifier performance and the accuracy of a thematic map
- The number of testing samples that should be used to test thematic map accuracy
- The number of testing samples needed to estimate class areas accurately

Slide 3.6.3 Recall from our work on machine learning methods that classifier performance is assessed by selecting a sample of testing pixels, previously unseen by the classifier; the algorithm is applied to that set to check the performance of the classifier.

In selecting the set of testing pixels, the analyst has to ensure that biases are not introduced by the fact that the areal sizes of the classes might be quite different. If care is not taken, more testing pixels in a random sample will come from large classes, whereas smaller classes will have fewer pixels with which to test the operation of the classifier. We will assume we have avoided such a bias in what we are now going to develop.

The set of testing pixels is often referred to as **reference data** or, less frequently now days, as ground truth. The same terms are generally applied to training data.

It is common to express the results of testing the classifier by setting up an **error matrix,** as on the next slide. In the past it was sometimes called a contingency matrix or a confusion matrix, terms you may still find in use.

Slide 3.6.4 Here we see an error matrix which has been complied from a thematic mapping exercise involving just three classes A, B and C. The cells are populated according to whether pixels have been placed into their correct class (assessed by comparing the classifier output with the reference data), or whether they have been put into the wrong class—that is, a classification error has been made.

The rows represent the classifier. For example, it put 39 pixels into class A; but when we look at what the reference data tells us for each of those pixels we see that some were **really** from class B and some from class C. Similarly, for its labelling of pixels as class B pixels and class C pixels.

Down the columns we see how each of the pixels in the reference data have been handled, by class. For example, in the first column the 50 pixels that are class A in the reference data have not all been put into that class by the classifier. Some have been put into class B and others into class C.

If there were no classification errors at all, the matrix would be diagonal, with zeros for all the off-diagonal entries. In such a case the classifier and reference data agree on the label for every pixel.

Slide 3.6.5 Here we emphasize that the row sums tell us about the classifier outputs. It has labelled 39 pixels as class A pixels, 50 as class B pixels and 47 as class C pixels. So that is how many pixels from each class we will find on the thematic map.

In contrast the column sums are the numbers of pixels in the reference data set for each class.

Slide 3.6.6 We now introduce some new nomenclature. In placing some class A pixels into the other classes, the classifier has committed errors. So, we refer to the off-diagonal entries along the same row as ***errors of commission***.

The errors down a column represent pixels from the reference data that the classifier has failed to label properly. We call those ***errors of omission***.

Slide 3.6.7 We now need to ask ourselves the question of how to determine the accuracy of a classification exercise using the entries in the error matrix. The answer actually depends on whether you are **the user** of the thematic map produced by the classifier, or **the analyst** who ran the classifier to label the pixels which led to the map. They are not the same.

A measure of how well the classifier has performed is how well it has labelled the reference pixels. If we choose class B as an example, the classifier was presented with 40 class B testing pixels and got 37 of them right—that is an accuracy of 92.5%. We call this the **producer's accuracy** since it is the accuracy with which the results were produced.

However, the user of the thematic map is interested in how many pixels labelled B on the map are correct. There are 50 class B pixels on the map, only 37 of which are correct. The others are the result of the classifier committing errors on class A and class C pixels. We call this the user's accuracy. For this example, it is 74%.

Slide 3.6.8 Here we see the full set of user's and producer's accuracies, along with a commonly used measure of classifier performance—the overall accuracy, computed as the total number of pixels labelled correctly by the classifier as a fraction of the total number of pixels in the image. Note the range of producer's and user's accuracies, compared with the average accuracy.

Slide 3.6.9 There is an alternative technique used for assessing the accuracy of a classifier, which does not used a separate set of testing or reference pixels. It is called cross validation, and involves the following steps:

- A single labelled set of reference pixels is used for both training and testing. It is divided into k separate, equally sized, subsets.
- One subset is put aside for accuracy testing and the classifier is trained on the pixels in the remaining $k-1$ sets.
- The process is repeated k times with each of the k subsets excluded in rotation.
- At the end of those k trials, k different checks of classification accuracy have been generated.
- The final classification accuracy is the average of the k trial outcomes.

A variation to this approach is to use subsets of size 1. It is then called the **leave one out** approach. There will be as many trials as there are reference pixels. Again, after all trials have been performed, we will have an estimate of the accuracy.

Slide 3.6.10

- Correctly assessing the accuracy of a thematic map is an essential step in operational remote sensing
- Accuracy assessment is carried out using labelled testing pixels—better referred to as reference data
- The error matrix summarises the performance information needed to assess class accuracy
- There is a difference between producer's accuracy (how well the classifier has performed) and user's accuracy (the accuracy of the thematic map produced by the classifier)
- Cross validation is an alternative means for assessing classifier performance
- The leave one out approach is a special case of cross validation often used in practice

Slide 3.6.11 These questions direct your attention to understanding fully the concepts involved in assessing classifiers and thematic maps. Pay particular attention to the third question about the cost overheads of the cross validation approach.

Lecture 7. Classifier performance and map accuracy

Slide 3.7.1 We now wish to extend our analysis of the accuracy of a thematic map generated by a machine learning classifier.

Slide 3.7.2 In the last lecture we looked at the difference between measures of the **performance of a classifier** class-by-class and **user's accuracy**, which is related to the accuracy of the thematic map produced by the classifier.

We now want to take this analysis further, by focusing on the true accuracy of the thematic map. We will show that user's accuracy only represents the accuracy of the thematic map if the selection of reference (testing) pixels by class reflects the actual ground proportions of the classes—in other words their prior probabilities.

To do this we need to use some simple probability theory.

Let the variates t, r and m represent the **true** labels for the pixels (in terms of what is actually on the ground), their labels in the **reference** data (data we have chosen to measure map accuracy) and their labels on the thematic **map** *produced by the classifier*. Hopefully the first two are the same, but we will treat them separately for a moment.

Let the two symbols Y and Z represent any of the available class labels. For our three-dimensional example they can be any of A, B or C.

Slide 3.7.3 Suppose we sample the thematic map that has been produced by the classifier. If we check the accuracy of those samples against their true labels on the ground, then effectively we are estimating the map accuracy probabilities

$$p(t = Z|m = Z)$$

This is the likelihood that the correct label for a pixel is Z if the map shows it as Z. That is what the user of the thematic map is interested in.

More generally, $p(t = Y|m = Z)$ is the probability that Y is the correct class if the map shows Z. This allows us to assess incorrectly as well as correctly labelled pixels.

If we take the reference pixels we have selected to assess accuracy and use them to check the corresponding map labels, then we are computing the classifier performance probabilities

$$p(m = Z|r = Z)$$

More generally, $p(m = Z|r = Y)$, which is the likelihood that the map label is Z for a pixel labelled as Y in the reference data.

Slide 3.7.4 If the pixels in the reference data set are in the same class proportions as on the ground, we can assume that the variates r and t are the same. That means **we have chosen our reference data set carefully so that the proportions of pixels in each of the classes are the same as their proportions in reality**.

We can then use r in place of t in our equations, so that, for example, the accuracy of the map from the previous slide is

$$p(r = Y|m = Z) \equiv \ p(t = Y|m = Z)$$

This is still the probability that the true label for the pixel on the ground is Y if the classifier (thematic map) says it is Z because we are assuming r and t are the same.

We can now use Bayes theorem to relate the accuracy of the map to the performance of the classifier

$$p(r = Y|m = Z) = \frac{p(m = Z|r = Y)p(r = Y)}{p(m = Z)}$$

$p(r = Y)$ represents the likelihood that class Y exists in the region being imaged—this is a property of the scene on the ground.
$p(m = Z)$ is the probability that class Z appears on the thematic map, and is given by

$$p(m = Z) = \sum\nolimits_{Y \in \{A,B,C\}} p(m = Z|r = Y)p(r = Y)$$

Slide 3.7.5 Let's take stock of where we are at this stage of the analysis:

The classifier labels pixels as m, whereas their labels in the reference data are r. Thus:

A classifier generates the probabilities that the classifier output will be Y if the reference data is Y

$$p(m = Y|r = Y)$$

whereas we are interested in the probability that the reference data is Y pixel if the map (classifier) says it is

$$p(r = Y|m = Y)$$

We now apply these concepts using our three-class example from the previous lecture.

Slide 3.7.6 Here we show the error matrix we are working with on the upper left-hand side of the slide. The table on the right, shows the classifier performance (producer's accuracies), the user's accuracies and the map accuracies computed from the expression

$$p(r = Y|m = Z) = \frac{p(m = Z|r = Y)p(r = Y)}{p(m = Z)}$$

with

$$p(m = Z) = \sum\nolimits_{Y \in \{A,B,C\}} p(m = Z|r = Y)p(r = Y)$$

If you wish to check those calculations, you will need to compute the full set of classifier performance probabilities $p(m = Z|r = Y)$ from the error matrix.

Note the effect of the class prior probabilities (that is, the real proportions of the classes on the ground) on the results. The map accuracies are only the same as the user's accuracies when the set of labelled pixels used for reference data is in the same proportion by class as the real situation on the ground.

The second and third examples show the differences if that is not the case. The last is an extreme but instructive example.

The average map accuracies are given by the formulas developed on the next slide.

Slide 3.7.7 Now consider the average accuracy of the thematic map generated by a classifier. Based on what we have done so far, the accuracy by class is expressed as the probability that the true class is Y if the thematic map says it is:

$$p(r = Y|m = Z)$$

In computing an average accuracy over all classes, we could be tempted just to average the above expression over all values of Z. However, that gives too much weight to the smaller classes and diminishes the contribution to the average of the performance on the larger classes

It is better, therefore, to compute the map accuracy as the average, weighted by the probabilities of occurrence in the thematic map:

$$\begin{aligned}\text{map accuracy} &= \textstyle\sum_{Z \epsilon \{A,B,C\}} p(r = Z|m = Z)p(m = Z)\\ &\equiv \textstyle\sum_{Z \epsilon \{A,B,C\}} p(m = Z|r = Z)p(r = Z)\end{aligned}$$

The last expression tells us that the true average map accuracy can be obtained from the classifier performance, provided the classifier performance probabilities are weighted by the prior probabilities. The average map accuracies noted in the final column of the previous slide are computed that way.

Slide 3.7.8 Some analysts prefer to express classifier performance in terms of the **Kappa coefficient**. It is a measure intended to avoid any 'chance agreement' between the performance of the classifier and the reference data. If we again use the results of the example from the last lecture, we can estimate the chance agreement.

The two sets of calculations on the right-hand side of this slide show the performance of the classifier by class, and the allocation of the pixels by class in the reference data.

The probability that they both place a pixel in the same class is given by the products of their individual performances.

The probability that both the classifier and the reference data place a pixel at random in the same class over all classes is the sum 0.106+0.108+0.117=0.331, which is the random chance of their agreeing on the label for a pixel.

On the other hand, the probability of a correct classification determined from the agreement of the classifier output and the reference data is (35+37+41)/136=0.831.

Slide 3.7.9 In terms of those calculations the kappa coefficient is defined as shown here, and for our example has the value 0.747. In order to give meaning to that number we use the ranges shown in the table here.

Slide 3.7.10 Kappa is not strongly supported by all analysts because its value is not a unique function of its arguments, as shown here.

Slide 3.7.11

- There is a difference between the performance of a classifier and the accuracy of the resulting thematic map
- Map accuracy can be obtained from classifier performance if the class prior probabilities are used
- Classification errors on large classes can give significant errors of commission into smaller classes
- The Kappa coefficient is a common alternative measure for expressing classifier performance
- Kappa suffers because its range of values does not naturally imply levels of performance, nevertheless it is often used as a comparative measure when jointly assessing different classification algorithms

Slide 3.7.12 In answering the first question you may wish to look at an extreme example. For example, consider a classification exercise with just two classes. Suppose in reality class A occupies 90% of the scene on the ground and class B the other 10%. Suppose your classifier is 100% accurate on class B, but has 50% error for class A pixels. What does that imply for the thematic map?

Lecture 8. Choosing testing pixels for assessing map accuracy

Slide 3.8.1 We need to consider one last topic on accuracy assessment to complete the theory aspects of remote sensing, before we move to radar.

Slide 3.8.2 When we looked at training classifier algorithms, we had to have enough training pixels per class to estimate reliably any parameters in the technique we chose.

We now turn to the related matter of choosing independent samples from a thematic map (reference or testing data) in order to get reliable estimates of its accuracy. If we don't choose enough then we will end up with a poor estimate of map accuracy.

Slide 3.8.3 We will undertake this analysis by looking at how well the testing samples we choose actually sample the thematic map.

In this slide, the top diagram represents the thematic map—we are going to work this theory out with just two classes but will comment on the multiclass situation later.

In a sense we don't have to know what happens with each of the two classes. All we are interested in is whether a pixel in the map is correctly labelled, irrespective of the class. By knowing how many pixels are correct overall, we can get the map accuracy.

In this top diagram the shaded pixels are those which have been labelled incorrectly by the classifier. We can describe the situation by a binary variate, which takes the value 1 if a pixel has the correct label and 0 if it is in error.

The second diagram shows a random sample of testing pixels. Because they are chosen randomly, and because we do not know which pixels in the map above are correct and which ones are in error, we have to choose enough testing pixels to be sure we sample the map so that our accuracy estimate is reliable.

The bottom diagram shows the testing samples laid over the thematic map. We define a new binary variate which takes the value 1 if a correctly labelled pixel is sampled and 0 otherwise.

Slide 3.8.4 First, we look at the distribution of errors, described by the variate y_i as shown. That variate has a binomial distribution. If we take the sum of all y_i we get the total number of correctly labelled pixels (the incorrect ones contribute 0 to the value of the sum), and the accuracy of the map can be expressed as

$P = \frac{1}{N}\sum_{i=1}^{N} y_i$ where N is the total number of pixels in the map

We are interested in getting an accurate estimate of P.

Slide 3.8.5 Now consider the properties of g_i. It also has a binomial distribution. Also, by definition $n < N$; otherwise we would be sampling every pixel in the thematic map!

Again, the sum $\sum_{i=1}^{n} g_i$ is the number of correctly labelled pixels—but in this case as found in the testing set. As before g_i is 0 for testing set pixels which are incorrectly labelled so they do not contribute to the value of the sum.

The proportion of correctly labelled testing set pixels, which is what we always use as the measure of map accuracy, is given by

$$p = \frac{1}{n}\sum_{i=1}^{n} g_i.$$

We want p, found from the testing set, to be a good estimate of the actual accuracy of the thematic map, P.

The question is, how many testing set pixels n are needed to ensure that?

Slide 3.8.6 Since the sum of binomials is itself a binomial distribution the map accuracy estimate p is binomially distributed. We can assume that g_i and y_i come from the same underlying distribution; they then have the same means, so that **expected value** of p is the equal to P

Slide 3.8.7 Although the *expected* value of the map accuracy found from the testing pixels is the mean P, the ***actual* value found can be different**. Hopefully, it would be in the vicinity of P but, depending on the number of testing samples taken, it might be quite different. How different might that be?

To answer that we look at the variance of p about the mean P which is given by the expression shown on this slide.

Usually the number of pixels in a remote sensing image far exceeds the number of testing pixels, so that the variance takes the form of the bottom equation on the slide.

It is inversely proportional to the number of testing pixels n, so a large n reduces the variance and makes the estimate p closer to the actual P.

Slide 3.8.8 Repeating the message from the previous slide—the variance reduces with more testing pixels. Thus, more testing pixels gets us closer to the true value of the accuracy of the thematic map.

To a good approximation p can be assumed to be normally distributed about the mean P, as illustrated in the diagram. Two standard deviations about the mean contain 95% of the population; thus with 95% confidence we can say that our estimate of the accuracy of the thematic map lies within about two standard deviations of the true value.

Let’s see how we can use that information.

Slide 3.8.9 With 95% confidence we know our estimate of the accuracy of the thematic map lies within about two standard deviations of the true value.

Now suppose we are happy for our estimate of the map accuracy to be within $\pm\varepsilon$ of its actual value, i.e. $p = P \pm \varepsilon$.

So, with 95% confidence we know it will be within the range if ε represents two standard deviations. Thus

$\varepsilon = 2\sqrt{\frac{P(1-P)}{n}}$ which gives the number of testing pixels required as $n = \frac{4P(1-P)}{\varepsilon^2}$.

Note that it is a function of the true accuracy P! Is that unusual? No. If P=1, in other words the map was 100% correct, we really don't need to sample it, whereas if P were low, the numerator $P - P^2$ is large, meaning that more samples are required if the accuracy is relatively poor.

Slide 3.8.10 As an example, suppose we are happy for an estimate to be within $\pm$0.04 of a true proportion which is thought to be about 0.85 (i.e. we are happy with the range 0.81–0.89); then at the 95% confidence level $n = 319$.

Randomly selecting 319 testing pixels will allow a thematic map accuracy of about 85% to be checked with an uncertainty of $\pm$4% with 95% confidence. Further examples are in the tables on the next slide.

Slide 3.8.11 Here we see explicitly that the required number of testing pixels decreases with the actual accuracy of the thematic map.

Slide 3.8.12 Our previous analysis was focused on the need to establish a sufficient number of testing pixels so that the **overall accuracy** of a thematic map could be assessed. What we want to know now is how to ensure we have enough testing pixels per class to know that **each class** is accurately labelled.

One approach, outlined in the paper referenced, leads to the numbers shown in the table below, in this case with an uncertainty in the estimate of $\pm 10\%$ at the 95% confidence level.

Slide 3.8.13 When looking at overall map accuracy we were able to use binomial statistics since only two outcomes are possible – a correctly labelled pixel or an incorrectly labelled pixel. If we wish to look at class-wise accuracies for each pixel in a thematic map a multinomial probability distribution is needed.

We will not go into the development here but note that it leads to the following conservatively high estimate for the number of testing pixels required per class,

When we are interested in whether we have the correct class for a pixel, and not just that it is correctly labelled, we can use the expression shown here.

ε is again the level of tolerance we need around the estimate of the class accuracy.

B is is the upper β percentile for the χ^2 distribution with one degree of freedom, where β is the overall precision needed, divided by the total number of classes. For the background theory see the paper by Tortora.

Slide 3.8.14 Consider this example from the book by Congalton and Green

Suppose we have 8 classes in a thematic map. We want to find the estimate their accuracies within a range of $\pm 5\%$ and we want our results to be at the 95% confidence level.

Then β = 0.05/8=0.00625 (0.625 percentile) of the distribution which gives B the value 7.568.

Thus, the total number of testing samples needed is $n = \frac{7.568}{4(0.05)^2} = 757$

Therefore, we need just under 100 per class.

Slide 3.8.15

- Just as in training, where we need enough labelled reference pixels to get good estimates of the parameters of a classifier, in testing we must have enough labelled reference (or testing) pixels to ensure we can get good estimates of the accuracy of a thematic map

- We need enough testing pixels to ensure that the **estimate** of map accuracy obtained is as close as possible to the **actual** accuracy of the thematic map generated by the classifier

- We can look for estimates of the overall map accuracy, or the accuracies with which each of the individual classes have been mapped. The latter requires more training pixels.

Slide 3.8.16 It is important to understand the messages behind each of these three questions if you are to be sure you understand the guidelines for selecting the number of testing pixels.

Lecture 9. Classification methodologies

Slide 3.9.1 Often, we see analysts somewhat naively using machine learning techniques for thematic mapping without thinking carefully about the objectives of the exercise, and whether a careful mix of procedures can produce better results. Here we want to look at some mixed algorithm methodologies.

Slide 3.9.2 Too often, analysts just apply a chosen classifier to an image, using labelled training data for each class, and expect good practical outcomes. Such an approach works acceptably for highly stylized images, such as the labelled data sets regularly used to test new algorithms, but they are not optimal when looking at real and heterogeneous image data.

In this lecture we will look at practical methodologies that work well in practice and yet are often overlooked.

We start with a review of the properties of supervised and unsupervised classification.

Slide 3.9.3 Remember, in **supervised classification** the analyst acquires beforehand a labeled set of reference data (training pixels) for each of the classes of interest and, desirably, for all apparent classes in the image. That data is used to estimate the parameters or other constants required to operate the chosen classification algorithm. The algorithm can then be applied to the full image of interest, and testing pixels can be used to assess how well it has performed.

The amount of training data will often be less than 1% to 5% of the image pixels. The learning phase, therefore, in which the analyst plays an important part in the labelling of pixels beforehand, is performed on a very small part of the image.

Once trained, the classifier can then attach labels to *all* the image pixels. That is where a significant benefit occurs in thematic mapping.

The output from supervised classification consists of a thematic map of class labels, often accompanied by a table of area estimates and, importantly, an *error matrix* which indicates by class the residual error, or the accuracy, of the final product.

Slide 3.9.4 In **unsupervised classification** clustering algorithms are used, typically on a sample of an image, to partition the spectral space into a number of discrete clusters or spectral classes. It then labels all image pixels as belonging to one of the spectral classes found, typically using a minimum distance assignment. A cluster map can be produced in which the pixels are given labels indicating which cluster they belong to.

Unsupervised classification is a segmentation of the spectral space in the absence of any information fed in by the analyst—that's why it is called unsupervised.

Analyst knowledge is used afterwards to attach information class labels to the map found by clustering, often guided by the spatial distribution of the labels shown in the cluster map. Clearly this is an advantage of unsupervised classification, as we will see in an example.

In general though, clustering algorithms are computationally expensive to run, compared with most supervised classification methods.

Slide 3.9.5 We can benefit thematic mapping by bringing together the strengths of supervised and unsupervised methods into a classification methodology. Even though very old, and devised when the maximum likelihood rule was the key machine learning classifier used in remote sensing, it teaches us a lot about operational thematic mapping. We will demonstrate the method with a simple example.

It uses the best aspects of the unsupervised and supervised approaches. Unsupervised clustering is used to discover the spectral classes in an image; and supervised classification, after having associated those spectral classes with the information classes of interest, produces the thematic map. It is outlined in 5 steps on the next slide.

Slide 3.9.6 The unsupervised-supervised methodology developed by Fleming and his colleagues works as follows:

Step 1
We use clustering to determine a set of clusters or spectral classes into which the image resolves. This is performed on a representative subset of data. Spectral class properties (statistics) can be produced from this step. Small classes that might be overlooked by the analyst, such as those consisting of mixtures along class boundaries, and elongated classes like rivers and stream systems, will be picked up.

Step 2
Using available reference data (maps, air photos, the image itself, the spatial distribution of clusters in the cluster map) we associate the spectral classes, or clusters, with information classes. Frequently there will be more than one spectral class for some information classes because they might look different spectrally to the sensor, even though the analyst calls them by the same name. This is an important consideration.

Step 3
We can then perform a feature selection to see whether all features (bands) need to be retained for reliable classification. Depending on the classifier algorithm used, and the data set, this step is not always necessary.

Step 4
We use the supervised algorithm to classify the entire image into the set of spectral classes. The hybrid approach was used initially with maximum likelihood classification, but it can be used with most classifier algorithms.

Step 5
Label each pixel in the classification as the information class corresponding to the spectral class and use independent testing data to determine the accuracy of the classified product.

Slide 3.9.7 We now use two simple examples to highlight the value of using unsupervised classification as a means for identifying spectral classes and for generating the signatures of classes for which the acquisition of training data would be difficult. Those results are then used in a supervised algorithm.

Slide 3.9.8 To demonstrate the hybrid approach, we use an image segment recorded by the HyVista HyMap sensor over the city of Perth in Western Australia. It is centred on a golf course.

The obvious cover types are water, grass (fairways), trees, bare ground including bunkers, a clubhouse, tracks and roads.

Apart from a few cases, the distribution of cover types suggests that it might be hard to generate training fields for all classes of interest.

Five bands were used: band 7 (visible green), band 15 (visible red), band 29 (near infrared), band 80 (mid infrared) and band 108 (mid infrared). The last two are the infrared maxima of the vegetation and soil curves, midway between the water absorption regions. It was felt that they would assist in discriminating among the bare ground and roadway classes.

The three heterogeneous fields shown on the image were used for clustering; between them they seem to include all cover types

The three fields were aggregated and the Isodata clustering algorithm was applied to that combined data set, with the goal of finding 10 spectral classes (there are 7 information classes). The results for the three cluster regions are shown on the top right-hand map. We can see that the spatial distribution of the clusters matches the spatial distribution of the information classes in the image, which is as to be expected, although there may be more than one cluster corresponding to each information class.

Slide 3.9.9 In this slide we show the mean vectors of each of the clusters found. By looking at where the clusters lie in the image and by looking at the shapes of the spectral reflectance characteristics seen by the distribution of the elements of the cluster means, we attach the information class labels shown.

Several important observations can be noted here:

First, two of the information classes are each represented by two spectral classes (clusters).

Secondly, one spectral class, or cluster, has been generated from water/vegetation mixed pixels along borders

Thirdly, elongated classes such as tracks and roads have been found as separate spectral classes. It would be a little hard to develop training pixels for those classes otherwise.

Slide 3.9.10 For verification, this slide shows an IR versus red bi-spectral plot for the clusters found. The distribution of classes agrees with where they would normally lie in such a plot.

Slide 3.9.11 Using the set of spectral classes found from clustering, and their mean vectors and covariance matrices which are generated automatically by the Isodata algorithm, we get the thematic map shown on the left-hand side of this slide. Tjis has been generated by maximum likelihood classification.

The right-hand thematic map is coloured such that the clusters, or spectral classes, corresponding to each information class have the same colour.

Slide 3.9.12 Here we show the final thematic map alongside the image, to allow a comparison. In this simple exercise we did not choose testing data so cannot report quantitatively on the map accuracy.

Slide 3.9.13 We now look at another simple example of the combined unsupervised/supervised approach.

It involves classifying an arid region of Australia employed for growing cotton by the use of irrigation from a nearby river. The task was to assess the area in hectares sown to cotton, as a surrogate for the amount of water used. Field agronomists had assessed the hectarage of cotton crops in the region but required corroborative evidence.

The image to be classified consists just of the visible red and the first of the two near infrared bands of a Landsat multispectral scanner image, recorded in February 1981. Although the region is very dry at that time of the year, apart from the crops there is a gallery (riparian) forest along the river, which provides another vegetation class.

Supervised classification was carried out using the minimum distance classifier—a very simple algorithm.

Slide 3.9.14 On the left of this slide we see a near infrared image of the region to be analysed. A test sub-image has been identified on which the results are to be evaluated. The Darling River can be seen in the image.

The cotton fields are mostly in the test area (white in the left-hand image, indicating high IR response), along with an approximately triangular shaped crop in the bottom south-eastern corner, and some other scattered fields.

The four rectangular selections in the right-hand image sub-set, along with a sample of the lower triangular crop, were used to resolve the spectral space into spectral classes by clustering. Here, the simple single pass clustering algorithm was used and each of the five heterogenous regions was clustered separately. The results are shown on the next slide.

Slide 3.9.15 The results of the clustering are shown here in a bi-spectral plot in terms of the means of the cluster centres found. There were 34 clusters in total, which were then rationalised down to the ten shown. That was done by associating the clusters with information classes using black and white and colour air photos, and photointerpretation of the image itself.

Those ten grouped spectral classes were considered adequate to differentiate the image into its main cover types and thereby avoid any errors of commission which might lead to poor estimates of the area of the cotton crops.

Slide 3.9.16 When the minimum distance classifier was applied to the test image, using the ten rationalized spectral classes from the previous slide, it was found that the cotton crops accounted for 803ha. The field agronomists had estimated 800ha.

It was not necessary to produce a thematic map in this exercise since the important result was the area of cotton in the test region.

Slide 3.9.17 How can we ensure we have pixels in the training data set representative of all the information classes in an image, and does it matter?

For example, consider the river/gallery forest class in the previous irrigated cotton crop example. How can the user select beforehand a representative set of those pixels with which to develop one of the binary classifiers in a support vector machine? If the pixels for that class are not well-differentiated and indeed may be mixtures of water/trees, then hand selection of the training fields may be difficult.

Again, the analyst may wish to consider using a hybrid clustering/SVM approach. The clustering step, as with the maximum likelihood and minimum distance classifier examples just presented, would be carried out on representative and heterogenous parts of the image in order to generate a set of spectral classes with which to work. Those clusters could be aggregated into single information classes before application of the SVM in order to limit the number of binary classifiers needed, although in some cases there may be value in retaining some information class sub-sets if that improves separability.

Slide 3.9.18

- Using unsupervised classification based on clustering can be a very effective way of generating training pixels and revealing spectral differences within information classes (spectral classes). Often, more accurate supervised results will be obtained if sets of sub-classes (spectral classes) are used for each information class

- Rather than use reference data explicitly to train a supervised algorithm, it is used to attach information class labels to clusters; this is a traditional unsupervised classification approach

- Even though the analyst might be interested in just a small number of classes (sometimes one) better results will often be obtained if all apparent cover types are represented in the classification, in order to avoid errors of commission caused by large classes spilling over into the smaller classes

- Because the convolutional neural network develops knowledge about a scene as a whole, and particularly the spatial nature of information classes, and implements exceedingly complex piecewise linear decision surfaces, it is unclear whether a prior unsupervised clustering will help improve performance at this stage, since very few real exercise have been performed.

Slide 3.9.19 Answering this question requires a careful assessment of the spectral reflectance characteristics of the cover types involved.

Lecture 10. Other interpretation methods

Slide 3.10.1 We don't always have to use supervised or unsupervised techniques for pixel identification in remote sensing. Other approaches used in practice.

Slide 3.10.2 Most of our analytical focus in the course ***has*** been on machine learning. There are however a number of other approaches used regularly in practice, often founded on scientific as against data analysis principles. Library searching using pixel spectra in hyperspectral images is one example of that we have already seen.

One of the long standing and effective methods for assessing cover types recorded in a remotely sensed image involve **indices** which are metrics computed from the brightness values of pixels in sets of bands. For example, a simple vegetation index might just be the ratio of a near IR measurement and a visible red measurement, the rationale for which becomes evident when we look at common spectral reflectance curves shown on the bottom left of the slide.

If we apply it to all the pixels in an image it will produce a map in which pixel brightness is correlated with a strong vegetation signature.

Slide 3.10.3 The indices should be defined as functions of surface reflectance, which is implied by our reference to the spectral reflectance curves. But in practice, the formulas are often based on the digital numbers (DN) in the received image product.

There are better vegetation indices, such as the two shown here. The most common is the normalized difference vegetation index—or NDVI.

Slide 3.10.4 Here we see an application of NDVI to NOAA AVHRR images, for which the pixel size is 1km. This allow us to develop a continental scale vegetation index map which demonstrates, in this case, the evolution of drought conditions over the past four years in Australia. The spring sequence is perhaps the most profound, showing the huge change in green biomass over the four-year period.

Slide 3.10.5 Many other indices are used in remote sensing, including those shown here which allow compensation for the amount of soil reflectance in a pixel, and those for monitoring water conditions, soil, green biomass, etc.

Slide 3.10.6

- Vegetation and other indices are the result of band arithmetic—i.e. computing new values for the brightness value of a pixel from sums, differences and quotients of the originally recorded bands; those new values emphasise properties of importance, such as greenness.

- Used in time series, indices can follow the development of important phenomena such as the development of a crop, or the evolution of a drought.

Slide 3.10.7 Answering this question requires you to plot the equations of straight lines that result from the index definitions. For example, from the simple vegetation index we have $NIR = VI * red$, where VI is the slope of a straight line through the origin.

Lecture 11. Fundamentals of imaging radar

Slide 3.11.1 We are now going to change direction completely and spend the rest of the course looking at the technology of imaging radar as a remote sensing tool.

Slide 3.11.2 In the first module of this course we looked at the Planck radiation law. Remember, that shows how bodies at different temperatures emit radiation or energy. As seen on the diagram to the right, the earth at 300° K is a low emitter by comparison to the sun or a burning fire.

But let's focus on the curve for the earth itself in more detail in the next slide.

Slide 3.11.3 Here we see the earth curve expanded; look particularly at the right-hand side covering the microwave range and note that the emission from the earth at microwave frequencies is much lower than its emission in the visible and infrared range.

Although there is a very small level of radiation from the earth at microwave frequencies we can for all intents and purposes assume the earth is dark. We can take advantage of that by irradiating the earth with an artificial source of microwave radiation just as we use a torch or flashlight at night when there is little natural light available.

Slide 3.11.4 When we use radar to image the earth's surface, we do so to the side of the platform whether that be satellite, an aircraft or a drone. The reasons for this will become clear soon.

There are two things we then need to understand. First, we need to understand how the energy reflected back to the platform is representative of the properties of the landscape. Secondly, we need to understand how such an arrangement gives us spatial resolution in both the across track and along track directions.

Just as with radio, television and mobile or cell phones, clouds are not a problem with radar imaging. This is a huge advantage because clouds can completely stop remote sensing imaging at optical wavelengths.

Also, since the platform carries its own source of irradiating energy, imaging can be carried out any time of day or night.

Slide 3.11.5 The energy radiated to the surface is not continuous but consists of a regular sequence of pulses as shown in this slide. Importantly those pulses travel at the speed of light which is 300Mms^{-1}. Reflections from targets closer to the platform will appear back at the radar earlier than those from targets further from the platform, as illustrated by A, B and C in the diagram.

The pulses are actually the envelopes of a burst of microwave radiation at the frequency or wavelength of interest. Just as with optical imaging, in which the earth responds differently at a different wavelengths, the same is true of radar. The frequency inside the pulse will determine the property of the earth that we are measuring.

Slide 3.11.6 The pulse transmitted in the previous slide is actually repeated on a regular basis—called the pulse repetition frequency. The rate at which the pulses are transmitted is synchronized with the forward velocity of the platform so that the strips of ground imaged by each of the pulses align with each other as shown in this diagram; this provides continuous coverage. It is actually the property of the antenna used to radiate the pulse that gives the so-called footprint on the ground.

The angle with which the radiation is transmitted to the side of the platform is called the **look angle**. Its corresponding version on the ground is called the **incidence angle**. They will be the same for a low flying platform and a flat surface. They will be slightly different, however, at spacecraft altitudes when the earth curvature might be important and for undulating terrain.

Slide 3.11.7 We now need to understand quantitatively how the radar separates targets, or regions on the ground, at different distances out from the platform.

In the diagram here we show two targets A and B separated in the direction out from the platform. We call that the ground range direction, in contrast to the slant direction, which is parallel to the beam from the radar, as shown.

Suppose the two targets are Δr apart in the slant range direction. Given that the radiation travels at the speed of light, c, and that it travels to and from the platform, the difference in time between the two echoes is given by $\Delta t = \frac{2\Delta r}{c}$.

If the pulse duration is τ as in the previous slide, then the two targets can be resolved provided they are no closer together than $\frac{c\tau}{2}$. We call that limit **the slant range resolution** because we cannot resolve between pulses if they overlap. That is illustrated on the bottom diagram of this slide.

Of course, as remote sensing users, we are interested in the resolution **along the ground range direction** rather than that in the slant direction. From trigonometry we can see that to be $r_g = \frac{c\tau}{2sin\theta}$.

Slide 3.11.8 From the last formula we can make some important observations:

1. There is no spatial resolution directly under the platform (θ=0). That is why the system has to be side looking. Some early aircraft radars of this type were called *side looking airborne radars* (SLAR).
2. The slant and ground range resolutions are independent of the altitude of the platform!
3. Ground range resolution is a function of incidence angle and thus will vary across the swath. It is best in the far swath where θ is largest, and poorest in the near swath where θ is smallest. This is opposite to that for optical sensors which have their best resolution closest to the platform, just as we can see more detail in the near range when we look out the window of an aircraft.
4. If the antenna radiated to both sides of the aircraft and a single receiver were used, then there would be a right-left ambiguity in the received signal. That can be circumvented using two antennas and receivers, but most often that is not the case. Most systems encountered in practice radiate to only one side of the platform.

Slide 3.11.9 Having looked at how we achieve resolution in the ground range direction out from the platform, we now need to see how we achieve spatial resolution along the track of the platform. Although a curious name it is called **azimuth resolution**.

It is determined by the properties of the antenna used on the platform for transmitting and receiving the microwave pulses. In particular, it is the so-called beam width of the antenna in the azimuth direction that sets the azimuth resolution, in the manner we now describe.

From antenna theory, the width of the beam transmitted by an antenna is directly proportional to the operating wavelength and inversely proportional to the length of the antenna, as shown on the slide. That means that a longer antenna will give a narrow beam width on the ground and thus a smaller strip irradiated as shown. What we want to know now is the width of that strip because that defines the azimuth resolution.

Slide 3.11.10 From the previous slide we have the top equation as the beam width of the antenna expressed in radians. At a range of R_o that projects onto the ground a dimension of r_a metres. That is the width of the illuminated strip and thus the azimuth resolution.

Note from this formula the azimuth resolution gets worse with increasing slant range and better with increasing antenna length. At this point it is worth doing a simple calculation, as on the next slide.

Slide 3.11.11 We consider an aircraft situation. The slant range is 2000m, the radar antenna is 3m long in the azimuth direction and the operating frequency is 10GHz. That corresponds to a wavelength of 3cm. Substituting those values into the formula, we see that an azimuth resolution of 20m results, which is acceptable. However, if the same radar is placed on a spacecraft where the slant range is 1000km, the azimuth resolution will be 10km, which is totally unacceptable.

Clearly, a better method is needed to achieve good azimuth resolution at spacecraft altitudes.

Slide 3.11.12 What if we used a short antenna and thus irradiated a broad region on the ground, as shown in the diagram? Clearly, the azimuth resolution should be very poor.
The length of the region illuminated in the along track direction is the slant range to the ground multiplied by the beam width of the small antenna, which is

$$L_a = R_o \frac{\lambda}{l_a}$$

What happens, though, if there is more than one pulse transmitted while the point target is in view of that large beam width?

To answer that, consider the geometry on the next slide.

Slide 3.11.13 Here we see the platform moving past the target. The radar just encounters the target when the target first comes into the beam and loses it when it leaves the beam. All the time the platform is transmitting pulses (which are often called **ranging pulses**) and receiving echoes.

By processing those echoes using a signal processing technique called matched filtering, we are able to make the length of spacecraft travel look like a long antenna (or "aperture"), as shown in the figure. That travel distance is equal to the projected beam width on the ground of the real short antenna.

The effective long antenna is called a synthetic aperture, leading to the name **synthetic aperture radar** (SAR) for the technology.

Slide 3.11.14 What beam width does the synthetic aperture create and what footprint does that lead to on the ground?

Remember the beam width, in radians, is the operating wavelength divided by the antenna length. For the synthetic antenna that is:

$$\Theta_s = \frac{\lambda}{2L_a} = \frac{l_a}{2R_o}$$

The "2" in the denominator comes about because the transmission is from the antenna to the target and then back to the antenna.

When projected on to the ground this gives an along track, or azimuth, resolution of

$$r_a = \frac{l_a}{2R_o} R_o = \frac{l_a}{2}$$

That is an amazing result because it tells us that the azimuth resolution in synthetic aperture radar is directly proportional to the antenna length in the azimuth direction. We will look at the implications of that in the next lecture.

Slide 3.11.15 By way of summary we note so far

- Imaging at microwave frequencies makes use of the fact that very little natural microwave energy emanates from the earth's surface; we can thus irradiate the surface with a source of microwave energy carried on a remote sensing platform

- As with optical remote sensing systems the forward motion of the platform sweeps out a swath of recorded image data of the earth's surface

- Imaging radars are side looking; range resolution is best at far swath (large incidence angles) and worst at near swath (small incidence angles)

- Good azimuth resolution is obtained by using a signal processing technique called matched filtering that allows a small real antenna to be used on the platform but achieves a very high spatial resolution equal to half the size of the real antenna; this is called synthetic aperture radar (SAR)

- In SAR both the ground range and azimuth resolutions are independent of the platform altitude and operating wavelength (which is the wavelength of the frequency burst modulated by the ranging pulse)

Slide 3.11.15 In the first question your attention is drawn to the need to receive sufficient energy that the brightness of a pixel is of a measurable value and well above any noise that might be present in the system. All electronic systems suffer noise and, whenever we do measurements, we are looking for signals that exceed the level of noise present.

Lecture 12. Summary of SAR and its practical implications

Slide 3.12.1 Let's now summarise where we were in the last lecture and then note some practical implications of synthetic aperture radar.

Slide 3.12.2 In this slide we summarise the important geometric properties of radar imaging, including how the width of the recorded swath depends upon the system parameters.

We know now how to resolve the landscape into pixels using radar. Our objective from this point on is to understand what the pixels tell us about the landscape.

On the way we have to consider a number of related matters, including some further properties of electromagnetic radiation, and sources of geometric and radiometric distortion in radar images.

Slide 3.12.3 The first important point to take note that of is that the waveform within the ranging pulse does not have a constant frequency but chirps over a limited range about a centre frequency. The range is very small but nevertheless it is a major determinant for achieving high range resolution.

Understanding that is beyond this series of lectures, but the book referenced provides the details for those interested.

The chirp range is small enough that we can regard the imaging wavelength as equivalent to that of the centre frequency.

Slide 3.12.4 A second practical matter, and one which has significant implications for understanding the properties of radar images, is to do with the polarization of the incident and reflected waves.

Power, or power density is carried forward as the result of both an electric field and a magnetic field that oscillate at right angles to each other and to the direction of propagation as illustrated by the field vectors in the left hand diagram; there is a fixed relationship between the magnitude of the magnetic field and the magnitude of the electric field, and they are always at right angles to each other; thus we only need to consider electric field from now on.

The direction in which the electric field vector points defines the plane of **polarisation**; depending on the transmitting antenna used at the platform the radiation incident at the earth's surface can be vertically or horizontally polarised.

Slide 3.12.5 The transmitted signal can either be vertically or horizontally polarized. Irrespective of the polarization of the incident radiation, the scattered signal can also have both horizontal and vertical components. Whether either or both of those components can be received again depends upon the properties of the receiving antenna.

Generally, the same antenna is used for transmission and reception so whether it can transmit and receive both polarizations will depend on its design. There are four possible radar configurations:

transmit H and receive H, which is called HH
transmit H and receive V, which is called VH
transmit V and receive H, which is called HV
transmit V and receive V, which is called VV

Note that the first in the double letter convention is the received polarization, while the second subscript is the transmitted polarization. Most systems are either HH or VV and are called **single polarization radars**, whereas more sophisticated radars accommodate all four configurations and are said to be **fully or quad-polarized radars**.

Slide 3.12.6 This slide shows a radar image example over the city of Darwin, Australia in two different polarisations—HH and VV. It was recorded by the ASAR C band imaging radar carried on the Envisat platform. Although much of the image seems similar in the two polarisations, there are some notable differences as indicated.

In many cases polarization is an important discriminator in radar imagery.

Slide 3.12.7 Here is another Envisat ASAR example, over the city of Kalgoorlie in Western Australia; this time one of the images is cross polarized as indicated. Again, note the differences between the like and crossed polarized responses.

Note the comment on the bottom left hand side of the slide. Monostatic radar means using the same antenna for transmission and reception. Later we will relax that requirement, but for monostatic systems, which are by far the most common, we assume the two cross polarized responses are identical.

Slide 3.12.8 In this example we see all four polarization configurations. It was recorded by the TerraSAR-X satellite program. The colour composite image was constructed by displaying the horizontally polarized image as red, the vertically polarized image as green, and the two cross polarized responses as blue after they have been added.

Slide 3.12.9 So, bringing all this together we see that operational radars in remote sensing are defined by three important system parameters: the operating wavelength, the polarization and the incidence angle or range of angles.

Ever since radar was developed during the Second World War the range of wavelengths has been described by a letter designation. It is not unique, but the frequency and wavelength ranges shown here are the most usual in remote sensing. Of this set L, C and X are the most often encountered.

You may wish to take note of the formula on the right-hand side which relates operating wavelength in metres with frequency in megahertz.

Slide 3.12.10 In this slide we see the technical characteristics of a number of satellite and aircraft radar remote sensing missions.

Slide.3.12.11

- The radar resolution cell (or pixel size in a displayed image product) is defined by the ground range resolution and the azimuth resolution
- The ranging pulses are chirps about the operating frequency (wavelength); although that requires some sophisticated signal processing to resolve targets in the range direction, better sensitivity to weak targets is possible
- The important parameters of an imaging radar are the wavelength, polarization, and look angle
- Operating wavelengths or frequencies are described by letter designations, reputedly chosen as a means for disguising the operating frequencies of aircraft detection radars during the Second World War.

Slide 3.12.12 These questions are designed to consolidate your understanding of the geometric properties of a radar image.

Lecture 13. The scattering coefficient

Slide 3.13.1 We now turn our attention to the properties of the earth itself that are measured by the various synthetic aperture radar programs.

Slide 3.13.2 In focusing on the properties of the earth's surface detected by imaging radars, we start by analyzing a more traditional radar situation in which we detect a point target. We will then generalize that to distributed "targets" such as crops, forests and other ground cover types of interest in remote sensing.

Consider the situation shown in the diagram, in which we start with the concept of an isotropic radiator on the left-hand side, which is a transmitter which radiates equally in all directions. A simple light bulb can be thought of as an approximation to an isotropic radiator.

In the next couple of slides, we will work through the details of this diagram and the accompanying calculations.

Slide 3.13.3 We look first at the transmission of radiation from the isotropic source to the target. The power transmitted P_t spreads out over the surface area of a sphere. At a distance R_o the power density over that spherical surface is given by this first equation, which is just the power transmitted divided by the surface area of the sphere.

When a real antenna is used, rather than an isotropic radiator, the power density towards the target is increased by the gain of the antenna. The gain is just a number that tells us how much better the antenna is in the preferred direction, compared with isotropic behavior. It is dimensionless.

Slide 3.13.4 Now we have to describe the properties of the target in such a way that we can use it in our calculations.

We describe it by an artificial area σ (m^2) called its **radar cross section**, which is defined as a cross-sectional area at right angles to the incoming beam, such that the area "intercepts" power from the incoming power density and **re-radiates it isotropically**. The power available for isotropic re-radiation is given by the first equation on this slide.

The power density then produced back at the platform, as a result of isotropic radiation from the target is as given by the second equation.

In order to describe the actual power received back at the radar platform after scattering from the target, we introduce a property of the receiving antenna, called its **aperture**, which is an area at right angles to the incoming beam of power density, and which extracts power from that beam. As seen in the last equation on this slide, the aperture has dimensions of area.

Slide 3.13.5 The aperture of an antenna is a property we use when describing its ability to extract power from an incoming wavefront of power density—that is, when receiving. An antenna as a receiver can also be described in terms of its receiver gain G_r which is related to aperture by the top expression on this slide.

If we use receiver gain rather than aperture, the equation for received power from the last slide becomes as shown here.

This is a famous equation, called the **radar range equation**. Note that the received power varies inversely **with the fourth power of range**!

It also shows explicitly, as should be expected, that the power level received at the radar is a linear function of the target property—the radar cross section.

Note that the equation has been derived on the basis of a point target rather than the distributed landscape, which we will consider shortly. But point targets do arise in radar remote sensing imaging. Any isolated strong reflector smaller than the size of a pixel can behave that way. Examples include houses, isolated trees and targets intentionally located in the scene to help calibrate the radar. We will see examples of those later.

Slide 3.13.6 We need now to see how we can modify the radar equation to account for the imaging of distributed regions such as natural grasslands, forests, crops, oceans and so on.

To do that we represent the landscape by a collection of infinitesimal elements as shown in the figure on this slide. Each has an effective area of ds and a radar cross section of $d\sigma$. On the average, the region has a radar cross section per unit area of $\frac{d\sigma}{ds}$. This is called the scattering coefficient—it is given the symbol σ^o and has units of m^2m^{-2}. That means, in principle, it is dimensionless, but it is important in practice to maintain the seemingly strange units of m^2m^{-2} as we will see later.

We now treat each of the infinitesimal resolution elements as through it were a single isolated point scatterer and use the radar range equation we derived previously, as in the equations shown here. We then integrate all those contributions over the resolution element to get the received power from the whole element, as shown.

Slide 3.13.7 The received power is a linear function of the scattering coefficient, as it was for radar cross section. This is important because it means that the measured received power is in proportion to the scattering coefficient of a surface.

The scattering coefficient is usually expressed in the logarithmic units of decibels and its value is related to a reference level, in this case a scattering coefficient of $1m^2m^{-2}$. Base 10 logarithms are used and it is common to refer to the scattering coefficient as “sigma nought”.

Radar cross section for point targets is also expressed in decibels, referenced to $1m^2$ as seen in the second equation.

The decibel notation is very useful in practice, not just because the scattering coefficient and radar cross section can vary over several orders of magnitude, but also because values can be computed easily, as we see on the next slide.

Slide 3.13.8 This slide shows a range of scattering coefficients expressed in normal and dB form, along with some sample calculations which show how the dB form can used to compute new values readily.

Slide 3.13.9 Because we have radars with different polarization configurations, we add subscripts to the scattering coefficient to indicate which transmit-receive configuration was used to measure it.

For quad- or fully-polarized radars the four scattering coefficients are usually brought together into a "sigma nought matrix".

Slide 3.13.10 In more advanced treatments of radar imaging we like to refer to the transmit-receive situation in terms of electric fields rather than powers and power densities.

Our derivations here were based on the concept of power transmission and reception, but as we saw when discussing polarisation, the actual signal is carried as a combination of propagating electric and magnetic fields.

Just as we expressed the power view in scattering coefficients, we can do the same with fields and, in particular, the electric fields involved. Rather than scattering coefficients though we then describe the scattering events in terms of the elements of the scattering matrix indicated here.

Incidentally, this matrix equation tells us why the second subscript on the matrix element here (and in all treatments) is the transmitted or incident polarisation and the first is the scatterered or received polarisation. If they were the other way around the matrix-vector multiplication would not work.

Slide 3.13.11

- The transmitted radar signal is treated by assuming isotropic transmission, multiplied by the gain of the transmitting antenna; gain is just a number which tells us how much better the antenna is compared with isotropic transmission in the direction of interest.

- The target is described by an area called its radar cross section which is defined by assuming that the power scattered from the target is isotropically re-radiated.

- Distributed targets are described by radar cross section per unit area, which we call the scattering coefficient.

- Both radar cross section and the scattering coefficient are polarization dependent.

- Both radar cross section and scattering coefficient are usually expressed in decibels (dB).

- We can also describe the scattering behaviour of a target in terms of electric fields rather than powers, in which case we define and use the scattering matrix of the target.

Slide 2.13.12 The first two questions here are important in understanding why the scattering coefficient of the landscape imaged by remote sensing radars is a function of look or incidence angle.

The last question shows you that radar imaging is based essentially on two inverse square laws, applied at the same time!

Lecture 14. Speckle and an introduction to scattering mechanisms

Slide 3.14.1 In this lecture we want to look at a major consideration in terms of the perceived quality of radar images—they often look speckled in appearance and lack the crisp quality of optical images.

Slide 3.14.2 In the next lecture we are going to look at the nature of the scattering coefficient for various earth surface cover types. Before that, though, we need to be aware of this unusual problem of speckle, how it arises and how it can be treated. That's what we want to do in this lecture, along with an overview of what is to follow.

Slide 3.14.3 As just noted, one of the most striking differences in the appearance of radar imagery compared to optical image data is its poor radiometric quality; that is caused by **speckle**. The figure here shows a portion from an image of a fairly homogeneous region in which speckle is obvious.

Slide 3.14.4 Speckle is a direct result of the fact that the incident energy is coherent. It is assumed to have a single frequency and the wavefront arrives at a pixel with a single phase. What does that mean?

Incident sunlight, that forms the basis of most of our optical remote sensing, is not coherent—it is a collection of energy at the different wavelengths corresponding to the wavelength range of the imaging band, and there is no relationship between the phases of the different components.

The phase angle of the sinusoid which represents a propagating electric field, is a relative quantity which describes its position in time. In the diagram on the left, the phase angle is described with respect to the time origin. More often though we talk about the **differences** in phase between two or more sinusoids, as shown in the diagram on the right.

Slide 3.14.5 A direct consequence of the phase difference between two or more sinusoids is **interference**. When they add "in phase" we get a sinusoid of twice the amplitude, as in the top right-hand diagram. When they add "out of phase" we get zero, as in the bottom right-hand diagram.

In between we get somewhere between those two extremes depending on the phase difference between the two sinusoids, as illustrated

Slide 3.14.6 How does that explain speckle?

Typically, a pixel will be composed of a set of a very large number of incremental scatterers— we saw that when we developed the radar equation. Their returns combine to give the resultant received signal for that pixel. Such a situation is illustrated in the diagram here.

The net brightness of the resolution element or pixel will be the result of all the incremental reflections interfering with each other.

An adjacent pixel will have a different set of interfering scatterers but will have about the same average brightness value if both pixels are of the same cover type.

Slide 3.14.7 If nothing is done to reduce the speckle in a recorded radar image the speckle level will be too high to allow sensible interpretation of the image. Before radar imagery is released for use, it has usually undergone some form of speckle reduction.

There are many ways to reduce the influence of speckle. Some quite sophisticated filters are used, but often just a simple averaging of several images of the same scene is used. But averaging will unfortunately reduce spatial resolution since averaging is smoothing.

However, when the radar is **designed** it is anticipated that averaging will be needed to reduce speckle, so the designed spatial resolution, usually in azimuth, is higher than needed. After averaging, the azimuth and range resolutions approximately match, as required by the user.

The number of signals averaged in the terminology of radar imaging, is called the number of “looks”.

On the next slide we see an example of speckle reduction by averaging.

Slide 3.14.8 This is a simple made-up example, but it illustrates the point that averaging will reduce the noise variance (speckle) while retaining the average value of the signal (the scattering coefficient). The four separate images have been generated from a distribution with a mean of 50.7 and a standard deviation of 50.05. When four “looks” are averaged the mean is preserved but the standard deviation has been halved. The speckle variance (which is equivalent to its noise power contribution) has been reduced by a factor of 4.

Slide 3.14.9 In the next lecture we start our treatment of radar scattering mechanisms—that is how different earth surface elements scatter energy in radar remote sensing, leading to the formation of an image.

Here we summarise the situation so that we have an overview of what is to come.

The most common scattering mechanisms that contribute to the formation of images in radar remote sensing are illustrated here. Unlike the case of optical remote sensing where, because of the very short wavelengths involved scattering mostly occurs from surfaces, in radar the situation is more complex.

Slide 2.14.10

- The actual radar signal is carried by electric and magnetic fields. We concentrate just on the electric fields since the accompanying magnetic fields are directly related to the electric fields.

- Electric fields are represented as sinewaves, since that is how they propagate; as a result, sets of electric fields can experience constructive and destructive interference.

- The backscattered signal from a pixel is composed of a large set of reflected fields from the myriad of scatterers that compose the pixel; those fields interfere with each other, the result of which is that adjacent pixels, even though from the same cover type, may have different brightnesses.

- That causes the image to have a speckled appearance; even though adjacent pixels might have quite different brightness values because of speckle, the average brightness (scattering coefficient) over a set of pixels of the same cover type will be representative of that cover type.

- Speckle can be reduced by averaging, which doesn't affect the average value of pixel brightness but reduces the variance, thus the image will look less noisy.

Slide 3.14.11 The last two questions here are included to add to your understanding about look averaging in radar remote sensing.

Lecture 15. Radar scattering from the earth's surface

Slide 3.15.1 In this next set of four lectures we will look at how radar energy scatters from the earth's surface.

Slide 3.15.2 This treatment is important in understanding radar images and, in many ways, is similar to the importance of spectral reflectance curves in optical remote sensing.

We will look at

- Surface scattering
- Volume scattering
- Strong (corner reflector scattering)
- Scattering from the surface of the ocean

Along the way we will discover some other interesting and unusual facts about how microwave energy interacts with the surface.

Slide 3.15.3 Consider a beam of microwave energy incident vertically on a surface as shown below. To analyse what happens we have to work in terms of electric fields, and we describe the properties of the surface in term of electrical quantities—the **dielectric constant** and sometimes its **conductivity**. Conductivity does not appear often, but dielectric constant is a material property that is central to radar imaging. The dielectric constant of air is 1, whereas most natural media have dielectric constants in the range of about 1 to 80 or so at radio wave froequencies.

The diagram shows three components of electric field: that incident on a surface, the component reflected from the surface and that transmitted into the surface medium.

Note the definition of reflection coefficient on the right-hand side of this slide **for vertical incidence**. It is determined exactly by the dielectric constant of the medium. We also define a power reflection coefficient, as shown, and a transmission coefficient. We don't use the latter very often in remote sensing.

Slide 3.15.4 If we consider dry soil at microwave frequencies $\varepsilon_r \approx 4$ so that $R \approx 0.11$ Thus only about 11% of the incident power is reflected. The remainder is transmitted into the surface. The surface would look quite dark in an image.

By comparison, for water at microwave frequencies the dielectric constant is much larger: $\varepsilon_r \approx 81$ so that $R \approx 0.64$ Thus about 64% of the incident power is reflected. The surface would have a light tone in a radar image at vertical incidence.

Water is a strong determinant of radar scattering because it has a major impact on the dielectric constant of a substance. The diagram on this slide shows its effect on the dielectric constant of soil. The imaginary part is related to the conductivity of the soil—which tells us how effectively it will conduct electricity.

Slide 3.15.5 Now consider oblique, rather than vertical, incidence, but still with a smooth surface. If a beam of microwave energy is obliquely incident to a surface as shown in the diagram the reflection coefficient becomes polarisation dependent. The two formulas given show that fact.

Reflection from a smooth surface is called **specular** since it is mirror-like. Because the reflected field is away from the incident direction (the radar) the surface appears black in imagery. Examples include calm water bodies (e.g. lakes) and smooth soil surfaces.

Slide 3.15.6 Now consider rough surfaces. Most real surfaces exhibit some form of roughness when being imaged by radar, as indicated in the figures here.

We now wish to understand the back-scattering properties of rough surfaces. However, how do we know beforehand whether we should consider a surface as rough?

We use the Rayleigh criterion shown here for that purpose. Note that the decision is related to wavelength, incidence angle and the variation in height of the surface.

Slide 3.15.7 In the previous slides we looked at scattering from smooth to approximately rough surfaces. Now what about the other extreme—a surface that is **very** rough? Such a surface is called **Lambertian**, for which the backscattering coefficient is as shown here. Notice that it is independent of polarization.

It is possible to derive scattering models for surfaces with roughness ranges between pure specular and Lambertian. Those derivations are beyond the scope of this course, but the results are shown on the next slide.

Slide 3.15.8 This slide shows three surface scattering models which indicate how the scattering coefficient varies as a function of incidence angle and surface roughness.

It is important to have a feel for these types of behaviour, particularly for how a smooth and very rough surface behave.

Slide 3.15.9 Surface scattering is sensitive to polarization. This graph illustrates the HH and VV like response of a moderately rough surface and the HV cross-polarized response; remember this means that the incident radiation was vertically polarized (as it was in the VV case) but there was some horizontally polarized scattering as well as the vertical component. When cross-polarized responses occur we sometimes say **depolarization** has occurred.

Slide 3.15.10 In this slide we summarise in two diagrams surface scattering behavior as a function of the common wavelengths found in radar remote sensing—as a function of surface roughness, and as a function of surface dielectric constant.

All wavelengths show strong dependence on the dielectric constant of the surface material; given that dielectric constant is a direct indicator of moisture content, this shows the importance of surface moisture in affecting the tone of a radar image; note that higher moisture gives a bigger radar response.

Variations in surface roughness show more strongly at the longer radar wavelengths; at X band it might be hard to discern such changes so the image will have a more uniform tone, irrespective of whether the roughness of a surface might vary across a scene.

Note that in general a rougher surface will give a bigger radar response.

Slide 3.15.11 This slide shows a classic radar image recorded by the short-lived Seasat satellite in 1978. The enhanced backscatter (lighter regions) result from increased soil moisture owing to the effect of a storm to the west, and subsequent storm cells that travelled to the north east, late on the day prior to image acquisition.

Slide 3.15.12

- The reflected field in radar for smooth surfaces is determined by the reflection coefficient of the air-surface interface.
- The reflection coefficient is a function of the surface material dielectric constant, which in turn is a strong function of moisture content.
- When viewed obliquely, as in a radar looking to the side, a smooth surface will appear black in imagery since there is little or no backscatter.
- As surface roughness increases the level of backscatter and thus the radar image tone increases.
- Backscatter from smooth surfaces is a strong function of incidence angle, whereas backscatter from rough surfaces is a weak function of incidence angle.
- Apart from specular scattering, backscattering from rough surfaces will have a cross-polarized (HV or VH) component, which is also a weak function of incidence angle.
- Longer radar wavelengths are more affected by surface roughness than shorter wavelengths.

Slide 3.15.12 The second question here starts you thinking about the choice of radar parameters for particular applications.

Lecture 16. Sub-surface imaging and volume scattering

Slide 3.16.1 We now want to look at the intriguing possibility of imaging below the surface with radar, along with the mechanism called volume scattering.

Slide 3.16.2 Not all of the energy incident on a surface is scattered. As seen in the last lecture some is transmitted across the boundary and into the surface medium. There it is absorbed by losses, usually very close to the surface. Sometimes, though, the losses are small enough to allow significant propagation into the medium, so that the energy is then scattered by buried features, allowing those features to be imaged.

The loss of power density with propagation into the surface medium can be described by a simple exponential equation, in which the exponent is called the **absorption coefficient**. It is given by the second formula on the slide, in which we see it is a function of wavelength, the dielectric constant of the medium and its conductivity, via the so-called imaginary part of the dielectric constant.

Slide 3.16.3 Based on those equations we can make a number of important observations.

If κ, the absorption coefficient, is large then power density drops quickly with distance into the medium.

κ is smaller for longer wavelengths indicating that there is better transmission into the surface material at longer wavelengths.

We now define the **penetration depth** δ as that value of r at which the power density has dropped to 1/e of its value just under the surface.

The diagram shows the penetration depth at L band as a function of moisture content. Note that for very dry media we can get penetrations of several metres.

It is important to note that the signal doesn't stop after the penetration depth—that is just a convenient measure of how quickly it falls off. Objects several penetration depths below the surface can still return measurable signals to the surface, noting that absorption happens on both the incident and reflected paths.

Slide 3.16.4 This is a wonderful example of the ability of spaceborne SAR to image below the surface of very dry sands. The top image is an optical colour infrared photo, and the bottom is a SIR-C composite radar image with the colour assignments shown.

The radar energy has penetrated below the sand sheet to reveal an old paleo channel of the Nile River, as shown. Adjacent buried drainage patterns are also visible in the bedrock under the sand sheet.

Slide 3.16.5 We now turn to the scattering of radar energy from volume media, such as tree canopies, shrubs and sea ice. These types of media contain many individual scattering sites that are hard to identify but which collectively contribute to a backscattered signal, as in the diagram shown on this slide.

A simple model of this type of behavior has been devised by assuming that the medium consists of a cloud of water droplets (scatterers). Known as the water cloud model, it gives the scattering coefficient of a volume medium as shown by the equation on the slide. Note that it is a function of the thickness of the canopy, and the volumetric properties of scattering coefficient and extinction coefficient. Note also that the loss of energy from the signal is the result of scattering by the fine particles of water.

Slide 3.16.6 We can use that model to simulate volume scattering behavior. The diagram here compares volume scattering with very rough surface scattering, in which we see that volume scattering is even less dependent on incidence angle than the roughest of surfaces.

Although many surfaces might show a small specular component near the origin for surface scattering (except for this very rough case), there is no specular component with volume scattering.

Slide 3.16.7 Here we make two important observations about volume scattering—one to do with polarization dependence and the other concerned with the frequency or wavelength dependence of the volume extinction coefficient.

The simple water cloud model of volume scattering is based on a set of small **spherical** scatterers. It therefore generates no cross polarized returns. In general, however, volume scattering is highly depolarising, which means it causes a cross as well as a co-polarised signal. That is because the scatterers are not spherical but have geometric shapes such as found with twigs and leaves.

As with like (co-) polarized behaviour (HH or VV), the cross polarized (HV or VH) backscattered signal is very insensitive to incidence angle but it is usually much lower in magnitude than the level of the co-polarised response.

The extinction coefficient for volume scattering is strongly dependent on wavelength, meaning that energy loss through the canopy is also wavelength dependent. If the canopy has low loss then imaging can be performed of the underlying medium, such as the soil surface and trunks underneath a tree canopy, and the water surface in the case of sea ice. In general tree canopy attenuation is

Almost negligible at	P band
Very low for	L band
Moderate at	C band
High for	X band

Slide 3.16.8 Although not a very good image, this JERS-1 image of a forested region in Australia shows how much higher the volume response of a forest is compared with the surrounding grassland—both are vegetated, but the grassland at L band is behaving more like a surface scatterer.

Slide 3.16.9

- Radar energy can penetrate very dry surfaces at long wavelengths.
- The absorption coefficient and the depth of penetration are strong functions of moisture content.
- Volume scattering is a weak function of incidence angle.
- In general, there are both cross and co-polarised components of volume scattering.
- Volume scattering is in general stronger than surface scattering at longer wavelengths.
- The canopy extinction coefficient for volume scattering is a strong function of wavelength.
- Canopies appear almost transparent at P band but as strong opaque scatterers at X band.

Slide 3.16.10 The second question concentrates on the relevance of penetration depth, while the first asks you to think about a scattering mechanism that we will meet in the next lecture.

Lecture 17. Scattering from hard targets

Slide 3.17.1 We now look at some unusual scattering behaviours that, nevertheless, occur often in radar imaging.

Slide 3.17.2 Because of the longer wavelengths involved, some materials we image with radar act as **hard targets**, in that they are capable of reflecting a very large portion of the incoming wave front and can appear very bright. Examples include

- Metallic elements such as wire fences (recall the forest example in the last lecture with the adjacent pasture fields)
- Facets directly aligned to the incoming radar beam; examples include roof facets
- Corner reflectors formed by paired horizontal surfaces and vertical structures; examples include houses, tree trunks and ships at sea

Because they are such strong reflectors, they tend to dominate the response of an individual radar resolution cell (or pixel), so that we revert to **radar cross section** to describe their scattering behavior, rather than the scattering coefficient.

We will now look at each of those in turn.

Slide 3.17.3 First consider simple metallic elements, such as wires appropriately aligned.

A wire fence, for example, can act as a strong scatterer. While the analysis is complicated, we can note that

- If the fence wire is aligned exactly at right angles to the incoming radar beam (which we call zero incidence angle in this context), and the electric field is in the plane of the wire, then it will reflect the beam strongly, exhibiting a high radar cross section.
- If the fence wire is of finite length, then it will also reflect strongly at angles just off zero incidence.
- If the electric field is not exactly aligned with the wire, there can still be some sizable reflection, particularly if the wire is thick.
- If the length of the wire, or another elongated metallic element, is a multiple of half a wavelength, the element is called resonant and will give an extremely high response if it is correctly aligned to the incoming beam.

Slide 3.17.4 Now let's look at facet reflection.

The roof of a building or another planar structure facing the incoming radar beam is called a facet scatterer. Its radar cross section is given by the formula shown on this slide.

We call this its **bistatic** radar cross section because the scattering (reflection) occurs at an angle 2θ away from the incoming beam.

Most of the time in radar remote sensing we are interested in the case where $\theta = 0$ so that the radar cross section becomes 4π times the square of the area of the facet, with the dimensions expressed as a fraction of a wavelength.

Slide 3.17.5 Now we come to a very interesting mechanism and one which occurs surprisingly frequently in practice in a number of different forms. It is a manifestation of dihedral corner reflector behavior.

The figure on the left shows a dihedral corner reflector; if it is large compared with a wavelength, then its radar cross section, within 3dB over 30-60 degrees, is given by the left-hand equation.

Now look at the right-hand figure, which is meant to represent the side of a building. The wall of the building has a radar image in the ground plane, but the length of that image will depend on the incidence angle, as indicated in the figure. Its radar cross section is therefore different from that of a simple dihedral corner reflector because length of the horizontal surface is not fixed.

If we have a standing structure such as a building, then the formula on right-hand side of this slide provides the appropriate model to use in order to understand its appearance in radar imagery.

Slide 3.17.6 We can use that same dihedral corner reflector approach as the model for a tree trunk in radar imaging. As shown on the left-hand figure here, the trunk has an image in the ground plane.

Theoretically, we could use the expression for the radar cross section of a cylinder to help develop a model for the trunk-ground combination. However, there is a simpler approach

If the trunk radius y is large compared with a wavelength, it is simpler mathematically to represent the cylinder as an effective flat plate, as indicated in the right-hand diagram, with the equivalent width shown. The radar cross section of the trunk standing on the surface is then given by the equation shown.

Slide 3.17.7 We need to make two modifications to the last expression to allow it to be used in real situations. First, for a tree, the vertical and horizontal materials are not ideal reflectors but are real substances such as wood and soil. We can account for that by introducing the power reflection coefficients for the trunk ρ_t^2 and the ground ρ_g^2 into the expression for the radar cross section.

Secondly, we have to account for the attenuation of any canopy that envelopes the trunk; we can do that by adding a two-way absorption term, giving as the final radar cross section of the tree the last formula on this slide.

We now have a complete model for the trunk in a forest situation and can use it to model the backscatter response of a forest in radar imaging.

Slide 3.17.8 The backscatter curves shown here were produced from the expression for a tree trunk acting as a corner reflector, as a function of canopy loss, with the parameters indicated. Note that this is the first time we have seen backscatter drop away at small incidence angles. That is because the reflection of the vertical structure in the ground plane shrinks away as the angle becomes smaller. The fall in backscatter at the high angles is the result of canopy attenuation, as can be appreciated when looking at the values of the extinction coefficient.

Slide 3.17.9 Because of the pathway involving the ground in trunk scattering, the radar cross section is a fairly strong function of the ground dielectric constant, as seen in the formula we derived for the radar cross section of a tree. It will change, therefore, with changes in the ground material, particularly if the ground changes from dry soil to water, as when a forest might be flooded. Remember, the dielectric constant at microwave frequencies of dry soil is around 5 or so, whereas for water it is about 81.

Therefore, a forest with a water understory will appear considerably brighter than one with a dry soil surface—as shown here it will be about 5dB brighter.

We see an example in the next slide.

Slide 3.17.10 This slide shows an image of a forest in Australia which sits across a river and which receives annual floods with snow melts further upstream. The bright response corresponds to flood water from the river under the forest as a result of the corner reflector effect.

In this study the change in backscatter from dry to flooded understory was found from the image to be about 6.8dB, compared with 5dB we might expect from the change in dielectric constant of the understory from dry soil to water. However, in this region the dry understory is unlikely to be sufficiently smooth as to be modelled fully by a surface reflection coefficient. Instead the surface may be more Lambertian, meaning less forward scatter in the dry double bounce model and thus a larger difference when flooding occurs.

Slide 3.17.11

- Strong scatterers include metallic structures appropriately aligned to the incident radar beam, along with facets and corner reflectors.
- In many cases strong scatterers will dominate the response of a radar resolution cell.
- Sides of houses, trees, ships at sea and ocean oil rigs will all give strong corner reflector responses.
- Trees without a dominant trunk will not behave as strong scatterers.
- Tree canopy attenuation has to be taken into account when modelling forests with strong scatterers, but not when considering buildings, ships and oil rigs.
- Changes in the ground surface can be assessed from radar imagery, even with a canopy, because of the change in ground power reflection coefficient.

Slide 3.17.12 The second question draws your attention to the equivalent of "canopy attenuation".

In the third question pay attention as well to cylindrical structures like tree trunks.

Lecture 18. The cardinal effect, Bragg scattering and scattering from the sea

Slide 3.18.1 We now look at some further unusual scattering behaviours, one of which is sea surface scattering.

Slide 3.18.2 This treatment will now finalise our consideration of scattering by looking at some unusual mechanisms that occur because of the coherent nature of the incident radiation:

- Bragg scattering, which occurs when there are periodic structures on the ground
- Sea surface scattering, which is related to Bragg scattering

Just before that though we want to look at another feature of the double bounce dihedral corner reflector model for buildings, particularly in urban and city regions where houses and buildings often occur in rows along streets laid out in a regular grid pattern.

Slide 3.18.3 In one of the quiz questions in the last lecture you were asked to consider what would happen to the response of a dihedral corner reflector if the incoming beam was not precisely aligned with the face of the reflector.

This is generally not a problem for tree trunks as a result of their cylindrical nature, but for flat vertical structures, such as the sides of buildings, if the incoming beam is inclined as illustrated in the diagram, the backscatter drops very quickly.

Thus, buildings will only stand out in an image if the direction of radar illumination is at right angles to the wall in the horizontal plane.

This leads to what is called the **cardinal effect**, an example of which is seen in the next slide. The word “cardinal” here derives from the cardinal points (N,S,E,W) on a compass.

Slide 3.18.4 Here we see a portion of a SIR-B image acquired over Montreal, Canada in 1984 demonstrating the cardinal effect.

The bright central portion of the image is where cross streets are aligned orthogonally to the incoming radar energy, whereas the darker portions to the north, of about the same urban density, have street patterns that are not orthogonal to the radar beam.

Slide 3.18.5 We now come to an unusual scattering mechanism, which had its origins in crystallography. The Australian father and son team of William and Lawrence Bragg received the Nobel Prize in 1915 for X ray diffraction, which has the same basis as what we are now to consider in radar scattering.

The wavelengths that are used in radar remote sensing (~3 to 50 cm) are not too different from some of the periodicities we see in the landscape, such as row crops

Consider the interaction of an incoming radar wave front with the surface shown in the diagram on this slide, which is periodic in the plane of irradiation. It could represent, for instance, the cross-section of a newly ploughed field.

When this region is irradiated there will be moderately strong reflections from regularly spaced parts of the surface as indicated. It is the *set* of those reflections which constitute the backscatter that occurs from the radar resolution cell (pixel).

Note that some returns travel further than others, so when looking at reconstructing the composite pixel response, we have to convert those additional travel distances into phase angles. Thus, the pixel response is made up of a set of signals with different phase angles; those phase differences will be related to the angle of incidence and the spatial wavelength of the surface periodicity.

Sometimes the signals will add constructively, giving a brighter than average return, whereas at other times they will add destructively. We saw the same effect when we considered speckle, but here the interference results from **periodicities** in the landscape, rather than **randomly distributed** incremental scatterers.

Slide 3.18.6 Usually when we consider the different mechanisms within a pixel which contribute to backscatter, we simply add the backscattered powers. With Bragg scattering, however, we add the electric fields, which then squares the backscattered power density, making the pixel much brighter than if the periodicity were not present

There is one important condition to all this, and that is that the row-like periodicity has to be aligned orthogonally to the incoming wave front; otherwise the phase reinforcement will not occur. When it does, the condition for constructive reinforcement is that the surface and electromagnetic wavelengths are related by the formula shown, which is called Bragg's law. The behaviour is called Bragg resonance.

Slide 3.18.7 Here we see an example of Bragg resonance with circular pivotal irrigated agricultural fields in Libya. The strong returns are most likely associated with Bragg Resonance; the radar illumination is from the bottom of the scene so that ploughed furrows running across the scene give the enhanced returns.

Slide 3.18.8 We now turn to how scattering occurs from the ocean. Our work on Bragg scattering is important here.

A flat sea will behave like a specular reflector and will appear dark in radar imagery. To receive measurable backscatter, the sea surface must be made rough by some physical mechanism. The principal means for surface roughening is the formation of waves. There are two broad types of wave, both excited by the action of wind blowing across the surface. They are distinguished by the mechanism that tries to restore the water surface against the driving effect of the wind.

Gravity waves depend on gravitation acting on the disturbed mass of water to counteract the effect of the wind; their wavelengths tend to be long, typically in excess of a few centimetres.

Capillary waves have wavelengths shorter than a few centimetres and rely on surface tension to work against the disturbance caused by wind action. They appear to ride on the gravity waves.

For both waves, amplitude and wavelength is a function of wind speed, fetch (the distance over which the wind is in contact with the surface of the water) and the duration of the wind event.

Slide 3.18.9 The bottom left diagram here shows the energy spectrum of typical capillary waves. It peaks around a wavenumber roughly aligned with C band radar. Importantly though, this diagram tells us that present among the capillary waves are components (in a Fourier analysis sense) that can match the wavelength of an incoming radar beam and thus cause Bragg resonance to occur.

From the spectrum we see there is more energy at smaller wave numbers; from the expression for Bragg resonance we see that for a given radar wavelength λ we can select a smaller wave number on the capillary wave spectrum if we make the incidence angle small. Thus, oceanographic radar imaging is usually best done with small incidence angles.

Slide 3.18.10 These images, recorded by Seasat and SIR-A over a region of the Californian coastline off Santa Barbara, illustrate the importance of incidence angle in sea surface scattering. The sea water detail is much better expressed in the 20 degree imagery than in the 40 degree imagery. Note, however, the terrain distortion at 20 degrees and the consequent difficulty in assessing terrain detail.

Note also the strong dihedral corner reflector responses form oil rigs in the channel, easily seen at 40 degrees but less so at 20 degrees.

Slide 3.18.11 Finally, notice what happens in radar images of the ocean if the capillary waves are damped. Oil slicks attenuate considerably the amplitudes of the capillary waves, meaning that incident radar energy has nothing to couple into, making those regions on an image dark, as seen here.

Note also the effect that ship wakes have on capillary waves, and thus on radar response.

Slide 3.18.10

- The dihedral corner reflector effect requires the incoming radar beam to be orthogonal to the reflecting elements.

- The cardinal effect shows how the alignment of street patterns affects strong reflections in urban zones.
- Bragg scattering occurs when the earth's surface has regular periodic features; it is also a strong reflection mechanism.
- Sea scattering entails coupling of the incoming electromagnetic radiation in the radar beam with capillary waves on the ocean's surface.
- It is strongest at small incidence angles.
- Oil slicks damp the capillary waves and thus considerably reduce radar backscatter from the sea surface.

Slide 3.18.11 When looking at the second question keep in mind that radar reflections are separated spatially in an image if their returns to the radar happen at different times.

Lecture 19. Geometric distortions in radar imagery

Slide 3.19.1 We now come to the last week of our course. In it we will look at some final topics in radar, including some innovative applications, and say something about how different remote sensing image data types can be used together.

First, though, we look at distortions in radar images.

Slide 3.19.2 The first form of distortion is shadowing. While not exactly a distortion, it does affect interpretation.

As with optical images, shadows are caused in radar when a signal cannot be received from behind an object. Unlike optical image shadows, however, shadowing in radar imaging is absolute.

For nadir viewing optical sensors it is possible sometimes to detect measurable signals from shadow zones because of atmospheric scattering of incident radiation into the shadow regions at (short) optical wavelengths.

Radar shadowing is likely to be most severe in the far range and for larger angles of incidence, whereas it is often non-existent for smaller incidence angles. That can be appreciated by looking at the diagram on this slide.

Slide 3.19.3 The first real geometric distortion we encounter is that to do with the changing range resolution across the swath. It is called **near range compressional distortion**.

Recall that the ground range resolution is best at far range and poorest at near range. Thus, the near range pixels cover a larger region of the surface than the far range pixels, yet both are made the same size in a display product, as illustrated in the diagram. As a result, near range features are compressed into smaller than realistic display pixels.

We see the impact of that on the next two slides.

Slide 3.19.4 Consider a region on the ground in which there is a square of grid-like feature such as field boundaries. Within each of the square cells there could be many pixels.

Imagine also that there are some diagonal lines as shown—they could be roads connecting across field corners.

The right-hand figure shows how that region on the ground will appear in recorded and displayed radar imagery. Not only do the near range features appear compressed but linear features at angles to the flight line appear curved. The combined effect is as if the image were rolled over on the near swath side.

Slide 3.19.5 The image at the top of this slide shows the effect again, but with a real image recorded by an aircraft radar. The bottom image has been corrected.

Slide 3.19.6 We now look at a distortion peculiar to radar imaging; it arises because objects are delineated in the range direction by differences in time delay.

As a result of that, how would a tall tower appear in the range direction in a radar image?

The radar echo from the top of the tower arrives back at the radar before that from the base because it travels a shorter two-way path. That causes the tower to lay over towards the radar on the image.

To emphasise that we draw concentric circles from the radar. All points lying on one of those circles will be at the same slant range from the radar and will create echoes with the same time delay. By projecting the circle which just touches the top of the tower onto the ground we see that the tower top, and indeed the whole tower, superimposes on ground features closer to the radar than the base of the tower. This effect is referred to as **layover.**

By comparison, in optical imagery vertical objects appear to lie away from the imaging device since they are superimposed on features further from the device than the base.

Slide 3.19.7 We now look at relief displacement, which is similar in origin to layover.

Instead of a tower, consider a vertical feature with some horizontal dimension, such as the model mountain shown in the diagram.

Using the same principle of concentric circles to project the vertical relief onto the horizontal ground plane, several effects are evident:

- the front slope is foreshortened
- the back slope is lengthened
- together these effects suggest that the top of the mountain is displaced towards the radar set

In ground range format they give the effect that the mountain is lying over towards the radar. If you go back to the Santa Barbara image in the last lecture that relief distortion is quite evident in the 20 degree Seasat image.

If we knew the local height, then the degree of relief displacement can be calculated. In principle, therefore, the availability of a digital terrain map for the region should allow relief displacement distortion to be corrected.

Slide 3.19.8 Not only is the relief displaced, but the brightness of an image is modulated by topography, particularly in mountainous regions. On front slopes the local angle of incidence will be smaller than expected and thus the slopes will appear brighter. On back slopes the angle of incidence will be larger than expected making them darker than would otherwise be the case.

Slide 3.19.9 This slide shows an example of relief displacement in a mountainous region, again with 20 degree Seasat radar imagery.

Slopes facing the radar illumination direction appear bright, while those away from the illumination appear darker, and there is rather severe terrain distortion evident in the small circle, most easily seen by comparing the top image with the optical image at the bottom.

Slide 3.19.10

- Because the formation of a radar image depends on resolving time delays in the range direction several unusual geometric effects happen which are not apparent in optical imagery:
    - The recorded image appears compressed at near range (as against far range for optical imagery)
    - Tall structures lay over towards the radar
    - Relief is displaced towards the radar

- The displayed brightness of an image is modified by relief, according to the scattering characteristics of a surface at different angles of incidence.

- If mountainous terrain were covered by a volume scatterer, such as shrubland at C band, relief modulation of brightness would be less evident.

Slide 3.19.11 The first three questions here should help consolidate your understanding of terrain distortions in radar imagery.

Lecture 20. Geometric distortions in radar imagery, cont.

Slide 3.20.1 With this lecture we finalise our consideration of distortions that occur in radar image data.

Slide 3.20.2 Let's start by looking at the material we have covered on geometric distortions in radar images so far. We can in fact make some general observations that are of value in choosing look (incidence) angles suited to particular purposes:

- For low relief regions larger look angles will emphasise topographic features through shadowing, making interpretation easier
- For regions of high relief larger look angles will minimise layover and relief distortion, but will exaggerate shadowing
- Relief distortion is worse for smaller look angles; low look angle missions such as Seasat designed for oceanographic applications often produce distorted imagery of high relief terrain. Recall the slide of ocean features in Lecture 18, which compared Seasat (20 degrees) with SIR-B (40 degrees) images
- From spacecraft altitudes, reasonable swath widths are obtained with mid-range look angles (35° - 50°) for which there is generally little layover and little shadowing. Look angles in this range are good for surface roughness interpretation

Slide 3.20.3 With this slide we introduce a new radar imaging concept.

Not unreasonably we have been concentrating on imagery that is representative of the earth's surface. That means in the across swath direction we have been concentrating on converting slant range resolution into ground range pixels; but that gives rise to strange features such as near-range compressional distortion.

Instead of producing images on the ground plane, we could keep them in the slant range format, as seen in this diagram.

In such a view the image has range coordinates measured along the slant direction rather than along the ground. A simple way to envisage the slant plane is to project it out to the side of the platform as shown.

An advantage of slant range imagery is that it doesn't suffer the near range compressional distortion encountered when the ground range form is used; this can be appreciated by looking at the series of full concentric rings in the figure, the distance between pairs of which represent the slant range resolution of the system. The dotted curves illustrate, however, that relief distortion occurs in both forms of imagery.

Slide 2.20.4 Let's now think about how we might correct geometric errors in radar imagery. There are two broad approaches to geometric error correction, depending on the level of topographic relief in the region being imaged.

For regions of low relief, in which shadowing and relief displacement distortions are not considered significant, near range compressional distortion can be removed readily because we can model its effect. There will be remaining errors in geometry caused by the same sorts of factors we saw with optical imagery—platform attitude and altitude variations, earth rotation, and so on—they are corrected by the same techniques. We will look at some aspects of that approach on the next slide.

For regions of high relief, we have to account for severe errors like relief displacement. If we had available a digital terrain map (DTM) of the region, it is possible to use it to model relief distortion since we know the mechanisms which give rise to it.

An artificial radar image of the region is created from the DTM, with shading added through the adoption of a surface scattering curve. The actual image is then registered to the artificial image (which has its pixels placed in the "distorted" positions). The relief distortion effects are then removed from the real, registered image by reversing the DTM distortions—the pixels have then been placed on their correct DTM positions.

Slide 3.20.5 When we looked at geometric distortion with optical images we used control points and mapping polynomials. We can do the same with radar images.

However, because of the presence of speckle, it is often difficult to locate naturally occurring control points to the required degree of accuracy. As a result, artificial control points are often created prior to recording the image data by deploying devices that will give recognisable returns in the received imagery. The positions of those devices are usually accurately known, through GPS fixing for example, so that image correction is assisted.

Those artificial control points can be passive or active. We will now look at both.

Slide 3.20.6 Although flat plates facing the incoming radar beam could be used to provide a bright facet-like reflection, they have to be aligned very accurately to ensure the maximum radar cross section is available. Their use as control points is therefore limited

Instead, trihedral corner reflectors are used. Having three faces they have a large range of angles around boresight over which their radar cross section is within 3dB of maximum.

For the square trihedral reflector $\sigma = 12\pi\frac{a^4}{\lambda^2}$ (23° beam angle)

For the triangular trihedral reflector $\sigma = 4\frac{\pi}{3}\frac{a^4}{\lambda^2}$ (40° beam angle)

Sometimes corner reflectors are deployed in set patterns so they are more easily recognizable in the recorded radar image.

Knowing the radar cross section of the reflector, it can also be used for radiometric calibration of an image.

Slide 3.20.7 Rather than relying on passive reflectors for control points and calibration, active radar calibrators can be used. They are simple transponders in that they have a receiver and a transmitter. The incident radar energy on reception is amplified by a given amount and then re-transmitted. As a result, the returned signal can be much larger than that from a passive corner reflector.

One small problem with an active transponder, which in radar remote sensing is called an **active radar calibrator** (ARC), is that the inbuilt electronics will introduce a small time delay in the reflection. That means the calibrator will appear at a larger range in an image than it really is. That is handled by making-up the time delay to a known amount, which thus gives a known range shift in the image, but which is easily corrected by subtraction.

A variation on the ARC is to have different polarisations for the receiver and transmitter antennas, so that cross polarized imagery can be calibrated. They are then called PARCs—**polarized active radar calibrators**.

Slide 3.20.8

- Shadowing in radar imagery is a problem in regions of high relief.
- The choice of incidence angle is significant in reducing shadowing, minimizing terrain distortion and highlighting sea surface features.
- Mid-range look angles are good for land surface remote sensing.
- Small look angles are good for sea surface applications.
- Radar images can be presented in ground range or slant range format.
- Control points for rectification of radar imagery can be synthesized using passive radar reflectors, such as triangular corner reflectors.
- Active radar calibrators can also be used for control points, and are also good for accurate radiometric calibration.

Slide 3.20.9 Does the second question remind you about our treatment of the cardinal effect?

Lecture 21. Radar interferometry

Slide 3.21.1 We now start a treatment of some innovative uses of radar imaging that are not available to us, in the same form, with optical imaging. But first we summarise the benefits of radar imaging.

Slide 3.21.2 As we have seen in this series of lectures on radar remote sensing, imaging with radar brings a number of advantages, including:

- Imaging can be carried out through cloud cover
- Imaging can be carried out at any time of day or night
- Sea state features can be detected and mapped
- Imaging is sensitive to soil moisture
- For very dry surfaces and long wavelengths, sub-surface imaging is possible
- Floods can be detected beneath forest canopies
- At long wavelengths forest backscatter is dominated by trunks, allowing woody biomass to be assessed
- At shorter wavelengths green leaf biomass can be assessed
- Radar scattering mechanisms are different from but complementary to those experienced with optical imagery, suggesting there are benefits in using the two imaging modalities together

Slide 3.21.3 Now let's look at a very different style of operation.

Because radar uses pure or coherent radiation for imaging, as against the incoherent sunlight used in optical imagery, we can interfere two radar signals intentionally to achieve a remarkable new application. Called **radar interferometry**, it allows topographic mapping to be carried out, and changes in topography with time to be detected.

We have seen interference in radar before, as in the speckle created by the interfering backscattered electric field vectors from incremental scatterers within a pixel, and in Bragg resonance. The relative phase angles of the fields cause the interference.

Those phenomena are natural scattering mechanisms that depend on phase differences. What we want to do now is use the difference in phase between two radar signals **deliberately**. We will see that that will allow us to determine the height of a pixel, as well as its latitude and longitude, and how they change with time.

The technique is commonly known as interferometric SAR, or **InSAR**.

Slide 3.21.4 Consider two radars operating side by side at the same altitude irradiating a region on the ground at height h above a datum. The horizontal distance between the radars is called the **baseline**.

Because the two radars are at different slant distances to the target (R_1 and R_2) there will be a difference in their phases on reception back at the radars for a pulse that is transmitted from both radars simultaneously. That two-way phase difference can be shown to be given by the first expression on this slide.

If we differentiate that phase difference formula with respect to the height of the region being irradiated, we get an expression for phase sensitivity which is a function of the known system parameters.

As an example, look at interfering two ERS-1 beams. Substituting in the formula shows us that we get almost 10 degrees of phase difference between the beams for every 1m change in elevation.

Slide 3.21.5 We can invert and integrate the last equation to get height as a function of the differential phase angle, in which α_{IF} is called the **interferometric phase factor.**

The constant of integration can be found from the height and phase difference at a particular location, if that is important. Thus, by finding the phase difference at each pixel position we can determine the corresponding pixel height, allowing a topographic map to be determined.

For completeness we note that we can represent the two separate radar signals at position (i, j) in terms of fields as shown in the centre of this slide. When they are received, the complex product is formed as shown, the angle of which is the interferometric phase difference. The total signal is called the **interferogram**.

Slide 3.21.6 There are some practical difficulties with using phase.

We saw that the phase difference between the received signals is

$$\Delta\varphi = \frac{4\pi B sin\theta}{\lambda},$$

which shows that it varies because of a change in incidence angle.

Two mechanisms affect incidence angle. The first is height variation, which is what we are interested in, and the other is position across the swath, which occurs even for a flat earth and effectively interferes with the height-dependent changes in incidence angle which we are trying to find.

This is called the flat earth variation of phase difference; it needs to be removed from the recorded phase difference between the two radars, which is reasonably straightforward to do; that correction is sometimes called **phase flattening.**

Slide 3.21.7 A second problem with phase, which needs to be compensated, concerns its cyclic nature. This can be seen in the example on this slide, which illustrates how there is an ambiguity in interpreting the difference in phase between two sinusoids. Phase angle is modulo 2π, so even though the absolute phase might surpass 2π in practice, mathematically it gets reset to the range (0, 2π).

Again, the effect can be compensated fairly easily before the image product—the interferogram—is produced. The process is called **phase unwrapping**.

Both phase flattening and phase unwrapping are standard operations carried out when generating topographic information from radar interferometry.

Slide 3.21.8 Interestingly, if we have two radars irradiating the surface there are two styles of operation.

The option exists for having both transmit and receive, or only one transmit and both receive. The latter style of operation, called the **standard mode**, is common when both radars are carried on the same platform. When two separate platforms are used, or a single platform is used on successive orbits with a baseline between them (called repeat pass interferometry), the former style of operation is used; it is called the **ping pong mode**.

Slide 3.21.9 Having summarized the means by which topographic detail can be mapped with radar interferometry we present in this slide an example produced by the TanDEM-X topographic satellite mapping mission. This was acquired using two satellites operating in the standard interferometric mode—one transmitting and both receiving. This map of Mt Etna is a very good example of the detail that can be obtained.

Slide 3.21.10

- Because SAR uses coherent radiation, which preserves the phase angle of the transmitted and received electric fields, it is possible to employ two radars side-by-side to detect terrain height.
- As a result, three-dimensional topographic maps can be created—usually for SAR they are called interferograms.
- Phase variations associated with the change of incidence angle, as though the earth were flat, have to be removed when creating the interferogram—that is called phase flattening.
- The 2π ambiguity in phase must also be compensated for—that is called phase unwrapping.
- There are two styles of operation—the standard mode and the ping pong mode.
- The two radars can be on the same platform, or on two platforms flying side by side.
- Alternatively, a single radar and platform can be used, but with images taken on subsequent orbits, which are then interfered to form the interferogram; that assumes that the landscape does not change during the repeat passes

Slide 3.21.11 These questions relate to three different aspects of SAR interferometry, all of which are important to the system designer but probably less so to the user of the interferometric product, which would be calibrated in terms of height.

Lecture 22. Radar interferometry for detecting change

Slide 3.22.1 Having looked at topographic mapping with SAR we now explore its use for detecting topographic change.

Slide 3.22.2 Suppose we now have two radars, one following the other in the same orbit (or subsequent orbits of the same platform) and that they image the same region of terrain at times 1 and 2 as indicated in the figure. Imagine that between those two times there has been a shift in a topographic feature in the range direction, as shown. That will show up as a shift in slant range, as indicated.

A two-way interferometric difference in phase will occur between the two acquisitions given by the formula shown here.

It depends only on the change in slant range as a fraction of a wavelength.

For ERS-1, for which λ=5.6cm, a (large) change in phase of 180^o corresponds to a change in slant range of 14mm! At θ=23^o that corresponds to a lateral shift (range direction) of 5.5mm. That is an amazing degree of sensitivity to change.

Slide 3.22.3 Ideally, there should be no horizontal separation of the two radars in along-track interferometry. If that is achieved then, from the work of our previous lecture, there can be no phase difference between the two returning signals resulting from unchanging terrain relief.

In practice, though, there will often be a horizontal baseline as well as a time separation (temporal baseline) between the radars. That means that the phase difference we are looking for, associated with topographic variations between the two times, can be obscured by the phase difference resulting from static topography; the latter has to be removed.

Removal of the phase difference associated with static topography can be done if a digital terrain model is available for the region. That can be used to synthesise, pixel by pixel, the topographic interferometric phase. That can be subtracted from the total phase difference on reception, leaving only the phase difference resulting from topographic movement. An alternative technique is to have three radar acquisitions, two of which are used to create an interferogram from the static topography (a DEM). Again, that is used to remove the effect of topography in the phase difference of two other acquisitions.

This latter technology goes by the name of **differential interferometric SAR** or DInSAR.

Slide 3.22.4 This slide shows three common methods for achieving the temporal baseline needed for along-track interferometry. One involves two antennas on the same platform, another entails using two passes of the same platform and the third uses two platforms in the same orbit, one following the other—a so-called tandem operation.

Slide 3.22.5 This is an early example of change detection using along track interferometry, based on two acquisitions by the ERS-1 satellite. Note the amazing sensitivity to the subsidence of Bologna.

Slide 3.22.6 This is a second example, in this case generated from JAXA ALOS-2 L band SAR data of the 2019 California earthquake. Again, note the enormous sensitivity to small changes in the landform.

Slide 3.22.7 And as a final example we return to the Mt Etna illustration. Now an acquisition by the TanDEM-X satellite has been used with a topographic map produced previously with the Shuttle Radar Topography Mission (SRTM) to produce a map of lava flows from an 8 April 2010 eruption of Mt Etna in Sicily. The SRTM used interferometric SAR to produce global scale topographic maps.

The SRTM DEM was acquired in February 2000, and the TanDEM-X acquisition was in October 2010. The SRTM data was used to remove phase variations due to static topography, allowing the phase changes associated with topographic change to be obtained from the TanDEM-X mission.

The left-hand image here shows differential phase variations associated with change. As seen the largest phase variations are those associated with lava flow. On the right-hand image the lava flow detected by along-track SAR interferometry has been overlaid on the topographic map (DEM) generated by standard cross-track SAR interferometry.

This is a great example of why SAR imaging is such an important and useful technology. Nothing like this can be so easily achieved with optical imaging.

Slide 3.22.8

- Topographic changes can be detected using along track interferometry (**ATI**).
- ATI can be achieved by mounting two radars on a single platform, such as an aircraft, by using repeat passes of the same platform, usually a spacecraft, or by flying platforms in tandem.
- When using temporal phase difference to detect topographic change, the phase difference between acquisitions resulting from any horizontal baseline must be removed.
- ATI is very sensitive to small changes in topography with time.
- See https://uavsar.jpl.nasa.gov/education/what-is-uavsar.html for some good recent examples of radar interferometry.

Slide 3.22.9 The last two questions here are two aspects of the same effect.

Lecture 23. Some other considerations in radar remote sensing

Slide 3.23.1 We now look at some other imaging modalities with radar imaging, expanding our understanding of the possibilities of this wonderful imaging technology.

Slide 3.23.2 The form of radar imaging we have been looking at so far is referred to as **strip mode**, because it produces a continuous strip of imagery, just as with optical remote sensing platforms. Other image types are also possible, three of which are illustrated in these next three slides.

Sometimes the swath widths generated in strip mode are not wide enough for some applications. The ScanSAR mode provides a wider swath, by breaking the imaging into cells in the across track and along track directions. The antenna used is steered electronically over the cells as shown. Each cell is treated as a mini swath in terms of the signal processing carried out to form an image. Composite swaths of several hundred kilometres then become possible.

The processing needed to produce a ScanSAR image is more complex than with conventional SAR because of the need to join the scanned cells, but this is manageable.

Slide 3.23.3 If the antenna beam does not look directly to broadside but, for example, looks forward, the system is said to have **squint**. That can happen if the platform yaws, in which case it is unintentional, or it can be the result of a planned maneuver. It leads to coupling of the range and azimuth dimensions of the image and to geometric distortion.

Slide 3.23.4 If the antenna is squinted forward and then steered backwards **as it passes and continues to irradiate** a target as seen in the left-hand diagram, then higher resolution of that target region is possible at the expense of resolution over the rest of the domain. This is called **spotlight** imaging.

The image of Sydney shown on the right is a great example of spotlight imaging, in which exceptionally good spatial resolution is achieved. We will see that in the next slide in which the region around the bridge is expanded.

The image was recorded by TerraSAR-X and created by averaging three high resolution spotlight images. The images were recorded on descending passes and the look direction was from the east (right in the image).

Slide 3.23.5 Here we see the former image zoomed in to the vicinity of the Sydney Harbour Bridge. Although there are three images of the bridge (is that unusual?) one is highly detailed. The form of construction is easily discerned, highlighting the exceptionally good spatial resolution achieved with the spotlight mode.

Take note of the other comments on the slide concerning the features of the image.

Slide 3.23.6 We now need to say something about the types of radar image product available. For those of you with an engineering or physics background this should be straightforward.

Radar image providers make data available in a range of formats, some tailored to particular applications. Apart from interferometric and tomographic products (see later), most will be derivatives of two fundamental formats: **single look complex** format and **scattering coefficient** format.

Remember that radar antennas radiate and receive electric fields, even though we often discuss the overall radar operation in terms of power and power density.

If the amplitude and phase angle of the received field is recorded, that is called a **complex signal**. It is also **single look** in the sense that multi-look processing to reduce speckle is carried out after reception. Taking these two properties together the data as recorded will be **single look complex**. Mathematically, after it has been processed to extract the ranging pulse data from the underlying carrier frequency, the received field signal can be described by the equation shown.

Slide 3.23.7 When providing the single look complex data product to the user, the radar operator converts the received field data into the complex elements of the scattering matrix S shown here, in which the operating polarization is indicated by subscripts, as we saw earlier.

Most often the user will acquire **scattering coefficient** imagery from the operator. Single look complex data is modified in two ways to produce that form. First, the field amplitude is squared to produce intensity, which is directly related to received power and, secondly, look averaging is performed to reduce speckle and to produce a resolution cell that is roughly square in shape. In the single look complex product, the resolution cells will often be highly rectangular, in anticipation of square cells resulting from look summing.

Again, in a multi-polarisation radar a matrix of scattering coefficients can be provided, as shown.

Slide 3.23.8 Here we see a variant of the traditional remote sensing imaging radar.

It is not strictly necessary to have the transmitter and receiver at the same location, as we have done up to now. Instead, we could put them at different positions (on different platforms). Such an arrangement is called **bistatic SAR** as against the **monostatic SAR** configuration we have been using so far. Clearly, there must be some form of communication between the transmitter and receiver so that the ranging pulse time from transmission to reception can be computed, but that is only a matter of a telecommunications link.

The transmitter could be on a space platform and the receiver on an aircraft. Also, the system can use “transmitters of opportunity”, such as the signals from navigation satellites, which reflect from earth surface features. Note that the sun is an energy source of opportunity for optical remote sensing.

Although the analysis is more complicated, shadowing effects are different for bistatic SAR, and earth surface features behave differently because just their **back**scatter is no longer important; instead, their scattering at angles different from the incidence angle become important.

Slide 3.23.9 We can generalize the bistatic concept.

It is possible to envisage a system with many transmitters and receivers. Variously called multistatic or network radars, their contemplated use is more common in surveillance applications than remote sensing but, with the trend towards clusters of small satellites for remote sensing applications, simple multistatic constellations are readily envisaged.

Note that each receiver in such a configuration receives scattered signals from the target as a result of every transmitter.

Slide 3.23.10 We now come to the next generalization of the operation of imaging radar.

Standard synthetic aperture radar generates images of the landscape in the two horizontal spatial dimensions, with detail in elevation projected onto that two-dimensional plane. In that respect it is similar to optical imaging.

InSAR, using two radars deployed across and/or along track, takes the next step and enables the mapping of topographic relief, and changes in relief with time, but it does not permit discrimination of detail vertically, such as the internal structure of a forest.

By appropriately utilising several radars, usually with vertical separation (as passes of the same radar platform on different orbits), it is possible to identify vertical structure with the technique known as SAR tomography, or sometimes **TomoSAR**.

Tomography resolves vertical detail by employing a synthesised *vertical* aperture much as azimuthal detail is resolved using aperture synthesis in normal SAR.

Slide 3.23.11 This slide shows a recent example of TomoSAR, showing the vertical resolution of forest detail. Five radar acquisitions, recorded by the DLR F-SAR system on 10 June 2014, were used to derive tomographic information. The F-SAR system is fully polarimetric and was operated at L band for this experiment. It can also operate at X, C, S and P bands.

The top left-hand picture shows a radar image; a white transect indicates where forest vertical detail will be mapped. The bottom left-hand picture is a lidar map of the canopy top, or ground, as appropriate. The canopy top information along the transect is seen on the very bottom right-hand picture. That allows a comparison with the canopy vertical (tomographic) detail shown in the top right-hand picture.

Slide 3.23.12 We have now completed our coverage of radar remote sensing. By way of summary of this last lecture note

- Wide radar swath widths can be achieved using the ScanSAR mode.
- Spotlight mode allows higher than usual resolution to be obtained over selected target regions.
- Bistatic radars use two platforms for imaging.
- Multistatic radar involves several transmitting and receiving systems. Each receiver can detect signals that result from scattering from each transmitter.
- Vertical detail can be resolved within a volumetric or compound scatterer using tomographic SAR.
- TomoSAR uses a baseline orthogonal to the direction of propagation.

Slide 3.23.13 The second question here is particularly important. It involves a scattering situation often encountered in radar imaging when multiple bounces of the incident radiation can occur before it is returned to the receiver.

Lecture 24. The course in review

Slide 3.24.1 We have come to the end of our three modules, in which we have looked at optical and radar modes of image acquisition in remote sensing, and their applications.

In this final lecture we look at the course in overview and consider what might be possible if we use more than one imaging technology in combination.

Slide 3.24.2 Our course covered:

- Sources of radiation for remote sensing
- The importance of the atmosphere
- Optical remote sensing imagery
  - Means for correcting errors
  - Means for analysis
- Machine learning methods
  - Feature selection
  - Sampling and accuracy assessment
- Radar remote sensing imagery
  - Distortions in radar imagery
  - Understanding radar scattering
  - Bistatic and multistatic radars
  - Interferometry and tomography

Now let's look at how we can use the different images types together.

Slide 3.24.3 Bringing different data sets together to extract information unable to be derived from a single source on its own is called **data fusion**. In remote sensing that term is often applied just to the registration step which ensures that the various image types are co-registered, and then registered to a map base. But in many fields data fusion refers to the **fusion of evidence** so that **joint decisions** can be made. We now wish to explore that interpretation briefly in remote sensing.

To see why that is important consider what we could do with registered optical and radar imagery.

One example might be to use machine learning methods to create a thematic map of land cover from optical imagery and then overlay that on a digital terrain model derived from radar interferometry, so that topographic influence on land cover can then be assessed visually.

Another example is to use thermal imagery to map temperature gradients in the water outflows from a power station while using optical data to map the surrounding land use.

In many cases such combinations of different data types can be done manually, but the more general case is interesting. For example, how can we devise machine learning techniques to combine the diagnostic capabilities of both radar and optical imagery to identify land covers or land use?

Slide 3.24.4 In this slide we look at some possibilities of what could be done using the two data types together to make joint inferences about cover types.

On the left-hand side we see the sorts of labels one might find with thematic mapping from optical data, along with those which might result from the analysis of radar imagery.

On the right-hand side we see what refinements of the label for a pixel might be possible if we use the two in combination.

For example, a pixel which exhibits the characteristics of vegetation in optical imagery, but behaves as a corner reflector in radar imagery, would most likely be a forested pixel.

Similarly, a vegetated pixel exhibiting Bragg scattering in radar might indicate row cropping, and so on …

How can we perform such combined inferences?

Slide 3.24.5 One approach is to fuse evidence at the label level, much as a human would do. The method for analyzing each individual data type would be matched to that data.

For example, a support vector classifier might be used to analyse a contributing hyperspectral image, whereas a knowledge of the incidence angle and polarization dependences of scattering types in radar might be used to analyse co-registered radar imagery. We can then use several different approaches to combine those recommendations, including sets of rules, such as the one illustrated. Such rules arise from the knowledge of expert image interpreters.

There are other approaches, including probably the use of convolutional neural networks. We still have to see concrete examples of how they can merge evidence from different data types to derive pixel labels. One prospect is to operate CNNs in tandem, the first layers to analyse of the individual data types and the later layers to do the fusion. Time will tell.

Slide 3.24.6

- Radar, optical and thermal imagery each provide quite different diagnostic tools for understanding ground covers (thematic classes).
- Optical imagery is determined by vegetation pigmentation, cellular structure and moisture content; by soil mineralogy and moisture content; by material suspended in water and by the bottom materials.
- Radar backscatter response is determined by the geometry of scattering elements on the earth's surface and by moisture content, via the property of dielectric constant.
- One way of forming joint inferences from the results of radar and optical image analysis is to make logical decisions based on user knowledge of how ground covers behave in both image modalities.

Slide 3.24.7 This question illustrates how expert systems, based on production rules, can be used to merge evidence from different data types.